

4MA1_2H_ch4_geometry_trigonometry

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1. 4MA1/2H_Winter_2025_Q26

A solid cone is joined to a solid hemisphere to make the solid shown below.

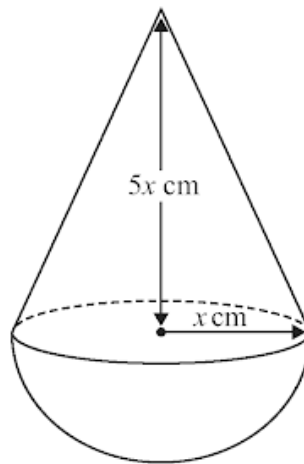


Diagram **NOT**
accurately drawn

The cone is made from copper.
The density of copper is 9 g/cm^3

The hemisphere is made from a different metal.

The total mass of the solid is 4752π grams
The total volume of the solid is $504\pi \text{ cm}^3$

Work out the density of the hemisphere.
Show your working clearly.

..... g/cm^3

6 marks

Shape **P** is similar to shape **Q**

The table shows some information about shape **P** and shape **Q**

	Surface area (cm ²)	Volume (cm ³)
Shape P	200	672
Shape Q	450	

Work out the volume of shape **Q**

..... cm³

3 marks

A, B, C and D are points on a circle with centre O

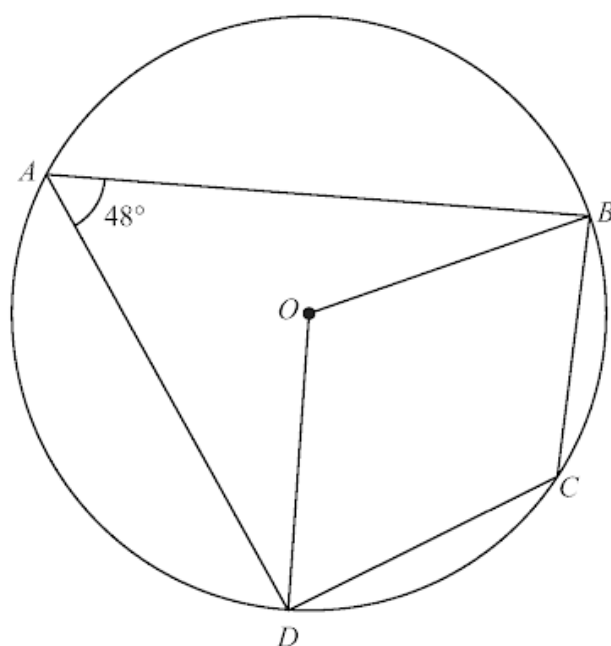


Diagram **NOT**
accurately drawn

Angle $DAB = 48^\circ$

(a) (i) Work out the size of the obtuse angle DOB

o

(1)

(ii) Give a reason for your answer.

(1)

(b) (i) Work out the size of angle BCD

o

(1)

(ii) Give a reason for your answer.

(1)

4. 4MA1/2H_Winter_2025_Q11

The diagram shows a right-angled triangle ABC

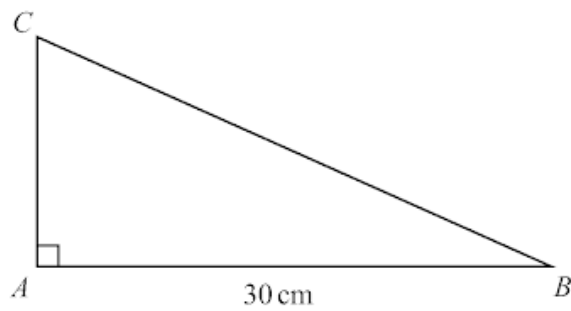


Diagram **NOT**
accurately drawn

Area of triangle $ABC = 240\text{ cm}^2$

Work out the perimeter of triangle ABC

..... cm

4 marks

5. 4MA1/2H_Winter_2025_Q6

A large circle with centre O contains 3 identical small circles.
Each circle touches two other circles.

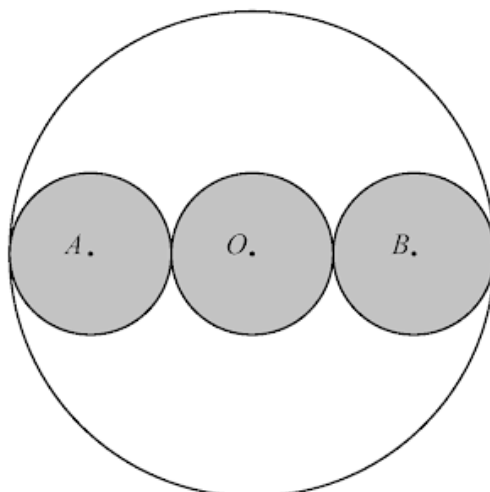


Diagram **NOT**
accurately drawn

A , O and B are the centres of the small circles.
 AOB is a straight line.

The circumference of the large circle is 160 cm

Work out the area of one small circle.
Give your answer correct to 3 significant figures.

..... cm^2

5 marks

Here are three similar quadrilaterals.

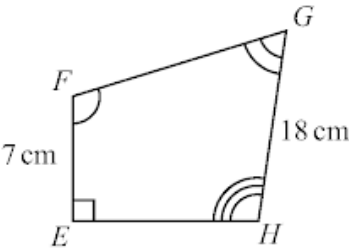
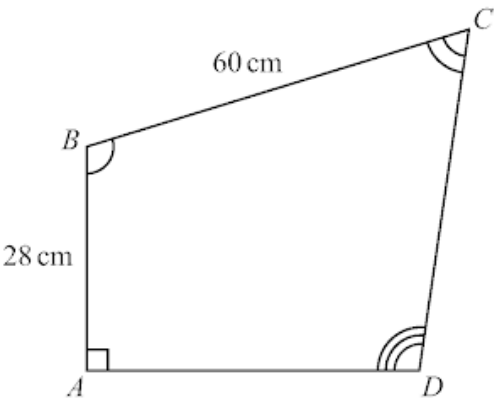
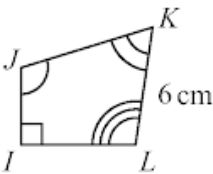


Diagram **NOT** accurately drawn



Work out the length of JK

..... cm

3 marks

7. 4MA1/2HR_Summer_2025_Q26

The diagram shows a solid hemisphere, H

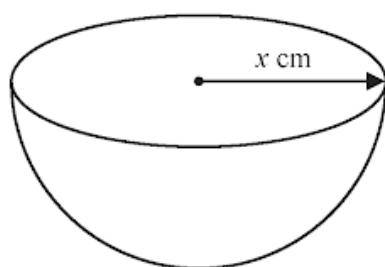


Diagram **NOT**
accurately drawn

The radius of H is $x \text{ cm}$

The volume of H is $6174\pi \text{ cm}^3$

A bowl is made by removing a solid hemisphere from H such that the uniform thickness of the bowl is 2 cm

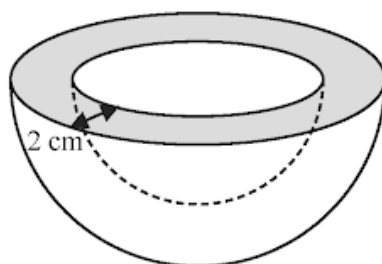


Diagram **NOT**
accurately drawn

Work out the **total** surface area of the bowl.

Give your answer in terms of π

..... cm^2

5 marks

8. 4MA1/2HR_Summer_2025_Q23

The diagram shows a triangular prism, $ABCDEF$, with a horizontal rectangular base $ABCD$

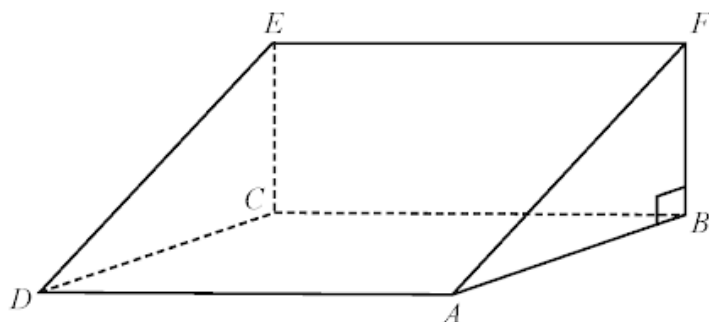


Diagram **NOT**
accurately drawn

M is the midpoint of the line AC

$AC = 40 \text{ cm}$ angle $CAE = 35^\circ$ angle $ABF = 90^\circ$

Work out the size of angle CME

Give your answer correct to 3 significant figures.

○

3 marks

A , B and C are points on a circle, centre O

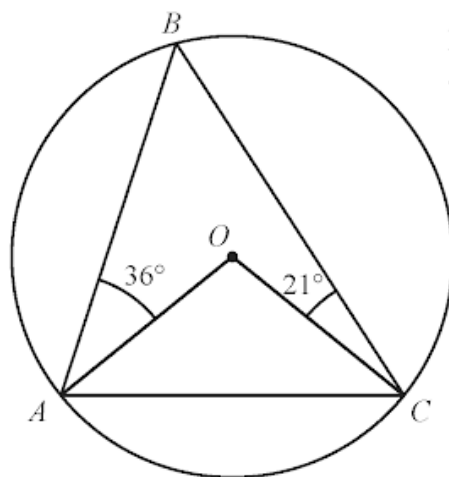


Diagram **NOT**
accurately drawn

Angle $BAO = 36^\circ$

Angle $BCO = 21^\circ$

Work out the size of angle ACO

o

3 marks

The diagram shows two triangles, ADE and CDB

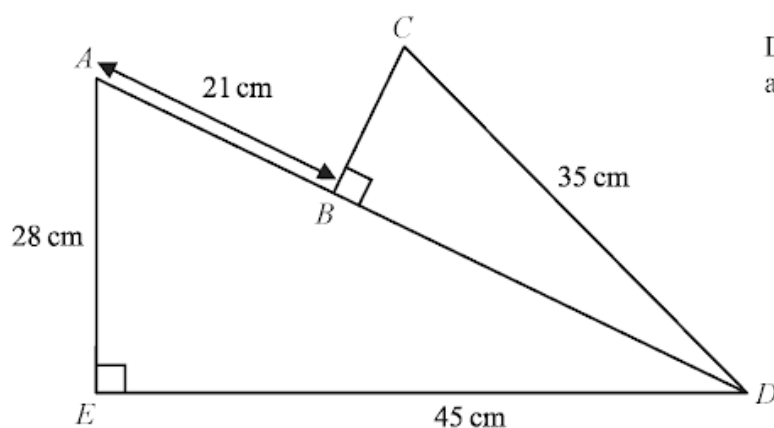


Diagram **NOT**
accurately drawn

ABD is a straight line.

$$AE = 28 \text{ cm} \quad ED = 45 \text{ cm} \quad AB = 21 \text{ cm} \quad CD = 35 \text{ cm}$$

$$\text{angle } AED = \text{angle } CBD = 90^\circ$$

Work out the area of triangle CDB

Give your answer correct to 3 significant figures.

..... cm^2

5 marks

A and **B** are two similar solids.

The height of solid **A** is 31 cm

The height of solid **B** is 18.6 cm

Given that

$$\text{volume of solid A} - \text{volume of solid B} = 735 \text{ cm}^3$$

find the volume of solid **A**

..... cm³

4 marks

The diagram shows a square-based pyramid $ABCDE$

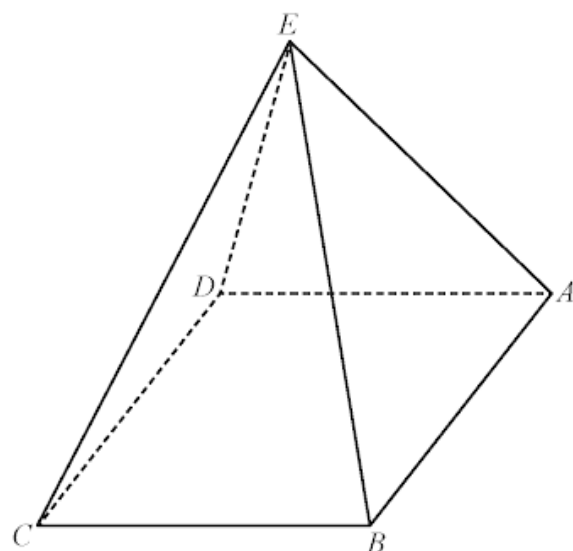


Diagram **NOT**
accurately drawn

$$EA = EB = EC = ED$$

M is the centre of the horizontal square base $ABCD$

Q is the midpoint of AB

$$\text{Angle } EQM = 80^\circ$$

$$EA : AB = n : 1$$

Find the value of n

Give your answer correct to 3 significant figures.

$$n = \dots\dots\dots$$

4 marks

The diagram shows a cuboid.

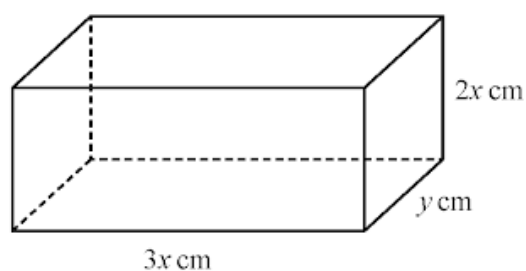


Diagram **NOT**
accurately drawn

The cuboid measures $3x$ cm by $2x$ cm by y cm

The volume of the cuboid is 1014 cm^3

The total surface area of the cuboid is $A\text{ cm}^2$

Show that $A = 12x^2 + \frac{1690}{x}$

You must show all the stages of your working.

3 marks

ABD and ABC are right-angled triangles.

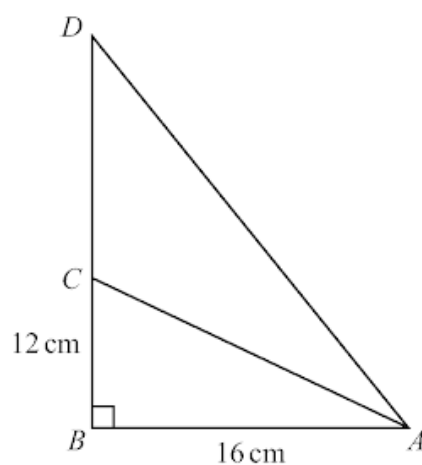


Diagram **NOT**
accurately drawn

$$AB = 16 \text{ cm} \quad BC = 12 \text{ cm} \quad AD = 1.5 \times AC$$

Find the length of CD

Give your answer correct to 3 significant figures.

..... cm

5 marks

The diagram shows two similar quadrilaterals, $ABCD$ and $EFGH$

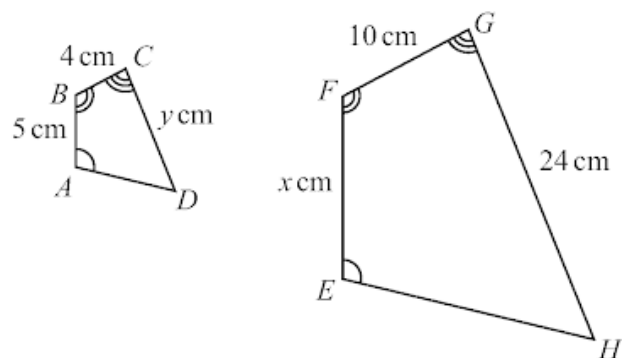


Diagram **NOT**
accurately drawn

$$AB = 5 \text{ cm} \quad BC = 4 \text{ cm} \quad CD = y \text{ cm}$$

$$EF = x \text{ cm} \quad FG = 10 \text{ cm} \quad GH = 24 \text{ cm}$$

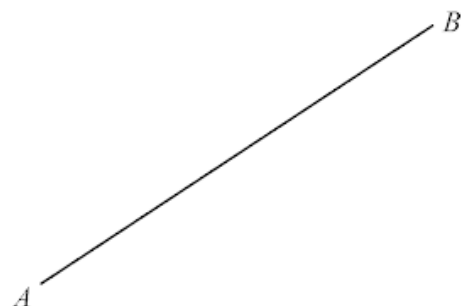
(a) Work out the value of x

$$x = \dots\dots\dots (2)$$

(b) Work out the value of y

$$y = \dots\dots\dots (2)$$

Using ruler and compasses only, construct the perpendicular bisector of the line AB
Show all your construction lines.



2 marks

The diagram shows an equilateral triangle ABC and a circle with centre O

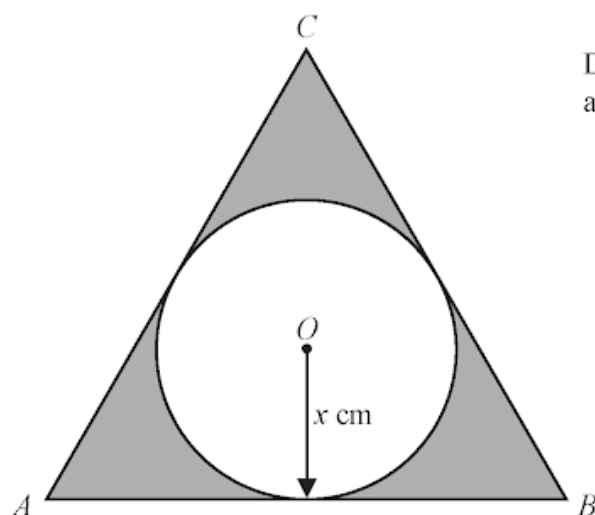


Diagram **NOT**
accurately drawn

AB , BC and CA are tangents to the circle.

The radius of the circle is $x \text{ cm}$

The total area, in cm^2 , of the regions shown shaded in the diagram is nx^2

Find the value of n

Give your answer correct to 3 significant figures.

5 marks

A , B and C are three points on horizontal ground.

$AB = 8.4$ metres $BC = 9.2$ metres

B is on a bearing of 067° from A

C is on a bearing of 129° from B

Calculate the bearing of A from C

Give your answer correct to the nearest degree.

6 marks

A, B, C and D are points on a circle, centre O

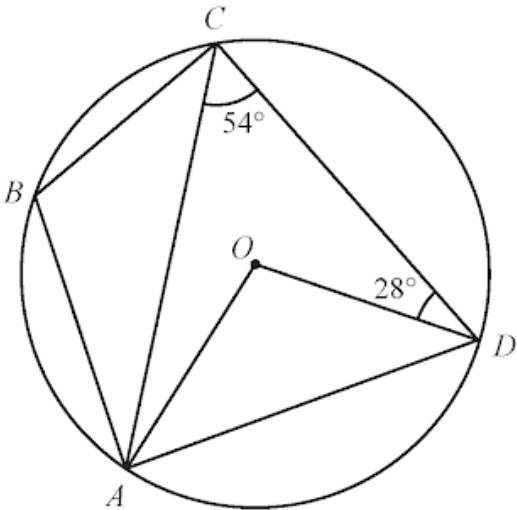


Diagram **NOT**
accurately drawn

(a) (i) Work out the size of angle AOD

.....
(1)

(ii) Give a reason for your answer to part (a)(i)

(1)

(b) Work out the size of angle CAO

.....
(1)

(c) Work out the size of angle ABC

.....
(2)

.....

.....

.....

.....

.....

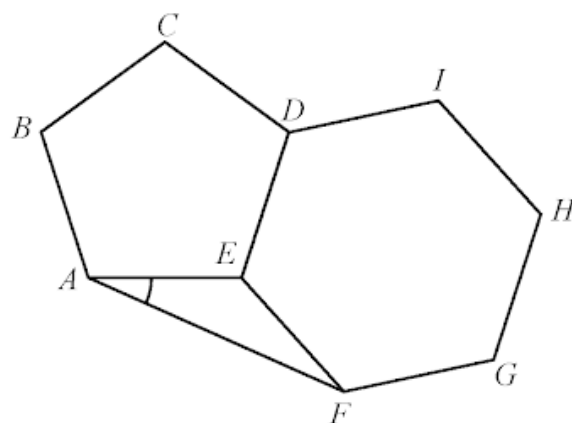


Diagram **NOT**
accurately drawn

$ABCDE$ is a regular pentagon.

$DEFGHI$ is a regular hexagon.

AF is a straight line.

Work out the size of angle EAF

5 marks

21. 4MA1/2H_Winter_2024_Q6

The diagram shows a solid triangular prism.

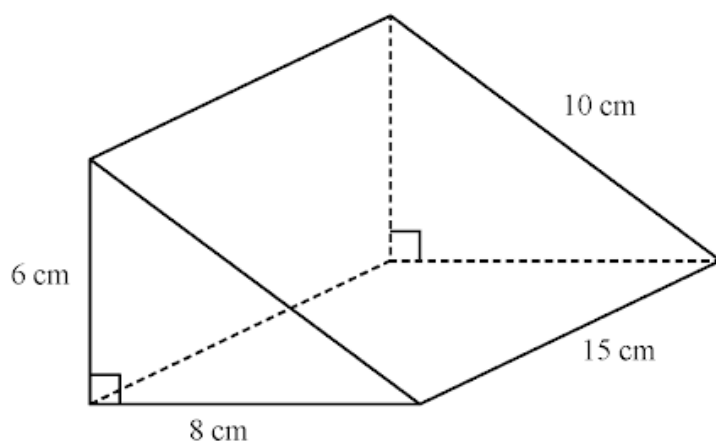


Diagram **NOT**
accurately drawn

Work out the **total** surface area of the triangular prism.

3 marks

22. 4MA1/2H_Winter_2024_Q3

The diagram shows triangle PQR

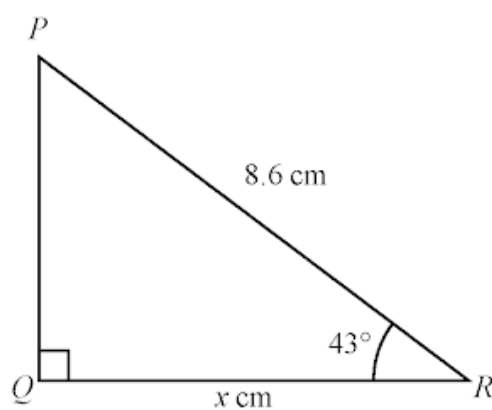


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to one decimal place.

3 marks

The diagram shows a solid, **S**, made from a cone and a hemisphere.

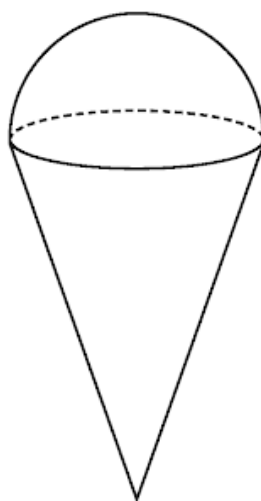


Diagram **NOT**
accurately drawn

The centre of the circular face of the cone coincides with the centre of the flat surface of the hemisphere.

The radius of the circular face of the cone, x cm, is equal to the radius of the hemisphere.

The total height of **S** is $4 \times$ the radius of the hemisphere.

A separate sphere has radius kx cm.

The volume of this sphere is $12.5 \times$ the volume of **S**

(a) Work out the value of k

(4)

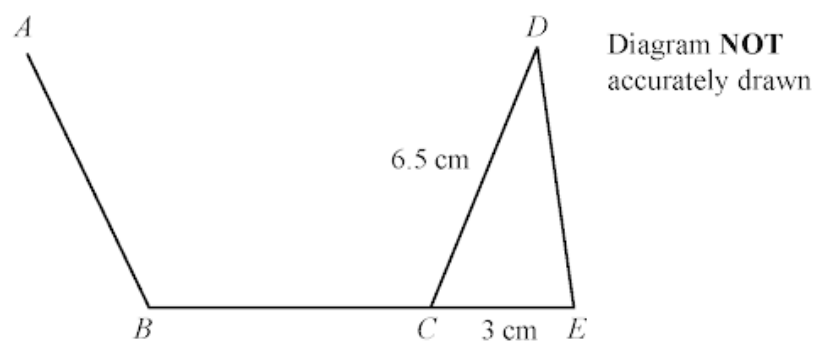
A solid, **T**, is similar to solid **S**

The volume of **T** is $512 \times$ the volume of **S**

The total surface area of **T** is $d \times$ the total surface area of **S**

(b) Find the value of d

(1)



AB , BC and CD are three sides of a regular pentagon and CDE is a triangle.
 BCE is a straight line.

$$CD = 6.5 \text{ cm} \quad CE = 3 \text{ cm}$$

Work out the area of triangle CDE
 Give your answer correct to 3 significant figures.

..... cm^2
 (3)

OAB is a sector of a circle with centre O and radius r cm.

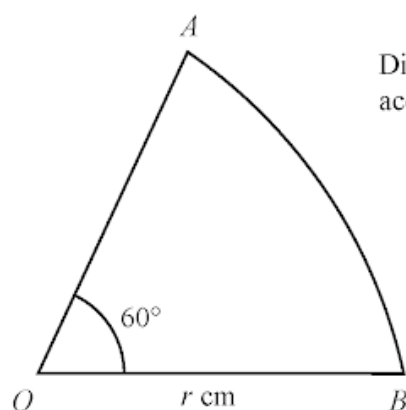


Diagram **NOT**
accurately drawn

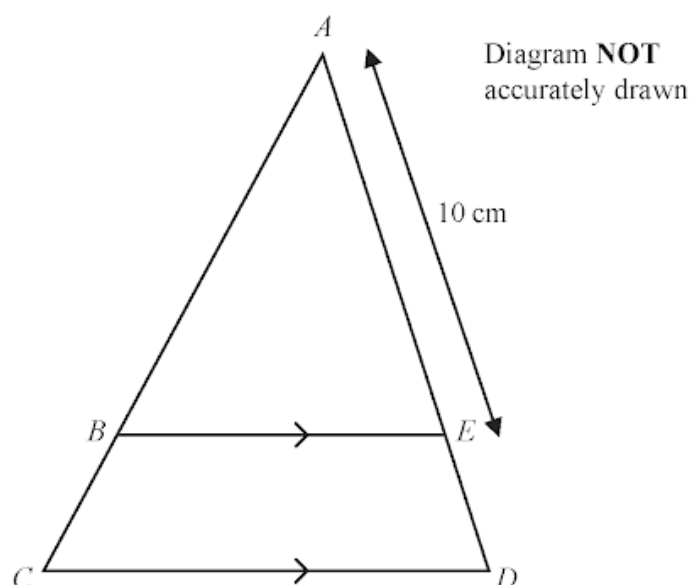
Angle $AOB = 60^\circ$

The perimeter of the sector is P cm.

Find a formula for P in terms of r

Give your answer in the form $P = r(c\pi + k)$ where c and k are values to be found.

(3)



In the diagram, ABC and AED are straight lines.
 BE is parallel to CD

$$AE = 10 \text{ cm} \quad \text{and} \quad CD = 1.5 \times BE$$

(a) Work out the length of ED

..... cm
 (2)

$$AB = (2x + 5) \text{ cm} \text{ and } BC = (3x - 5) \text{ cm}$$

(b) Work out the value of x

(2)

The diagram shows a hexagon $ABCDEF$

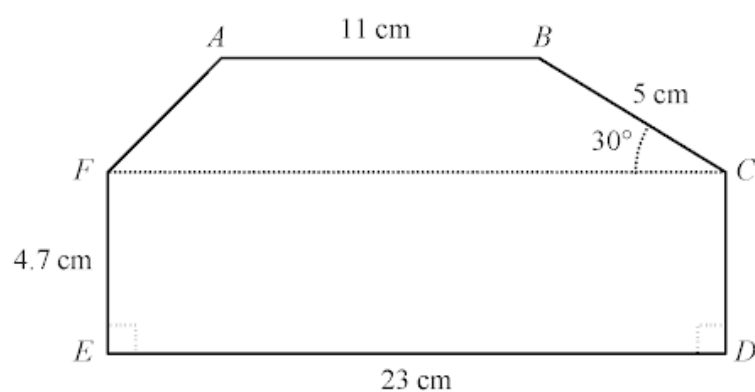


Diagram **NOT**
accurately drawn

Angle $BCF = 30^\circ$

AB , FC and ED are parallel.

Calculate the area of $ABCDEF$

Show your working clearly.

..... cm^2

(5)

The diagram shows the design for a badge, which will be made using wire.
The design is a circle inside a square $ABCD$

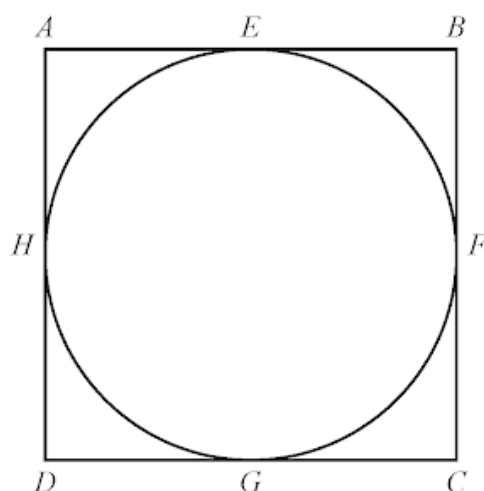


Diagram **NOT**
accurately drawn

The circle touches the square at the points E , F , G and H

The area of the square is 81 cm^2

Calculate the total length of wire that will be needed to make the square and the circle.
Give your answer correct to 3 significant figures.

..... cm

(4)

The diagram shows a square $ABCD$ and a circle.

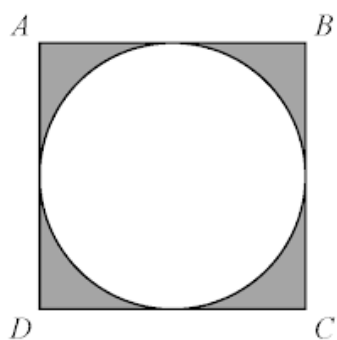


Diagram **NOT**
accurately drawn

The sides of the square are tangents to the circle.

The total area of the shaded regions is 80 cm^2

Work out the length of AC

Give your answer correct to 3 significant figures.

..... cm

(5)

30. 4MA1/2H_Summer_2024_Q7

The diagram shows a 6-sided shape $ABCDEF$

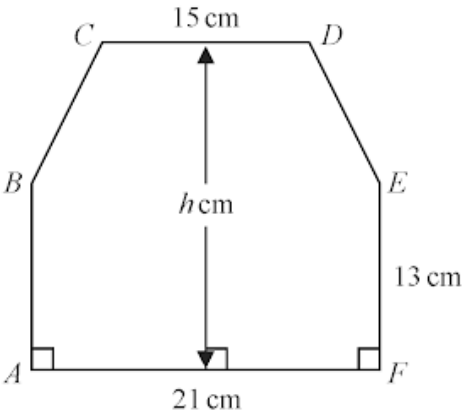


Diagram **NOT** accurately drawn

$AF = 21\text{ cm}$ $CD = 15\text{ cm}$ $AB = FE = 13\text{ cm}$

The perpendicular height of the shape is $h\text{ cm}$
 CD is parallel to AF

The area of the shape is 390 cm^2

Work out the value of h

(4)

31. 4MA1/2H_Summer_2024_Q5

The diagram shows triangle ABC

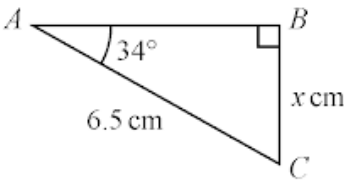


Diagram **NOT** accurately drawn

Work out the value of x
Give your answer correct to one decimal place.

(3)

$ABCD$ is a trapezium with AB parallel to DC

A is the point with coordinates $(-4, 6)$

B is the point with coordinates $(2, 3)$

D is the point with coordinates $(-1, 8)$

The trapezium has one line of symmetry.

The line of symmetry intersects CD at the point E

Work out the coordinates of the point E

The diagram shows a solid cone and a solid sphere.

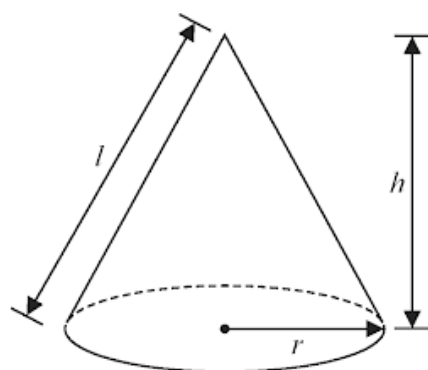


Diagram **NOT**
accurately drawn

The cone has base radius r , slant height l and perpendicular height h

The sphere has radius r

The base radius of the cone is equal to the radius of the sphere.

Given that

$$k \times \text{volume of the cone} = \text{volume of the sphere}$$

show that the **total** surface area of the cone can be written in the form

$$\pi r^2 \left(\frac{k + \sqrt{k^2 + a}}{k} \right)$$

where a is a constant to be found.

AEC and BED are chords of a circle.

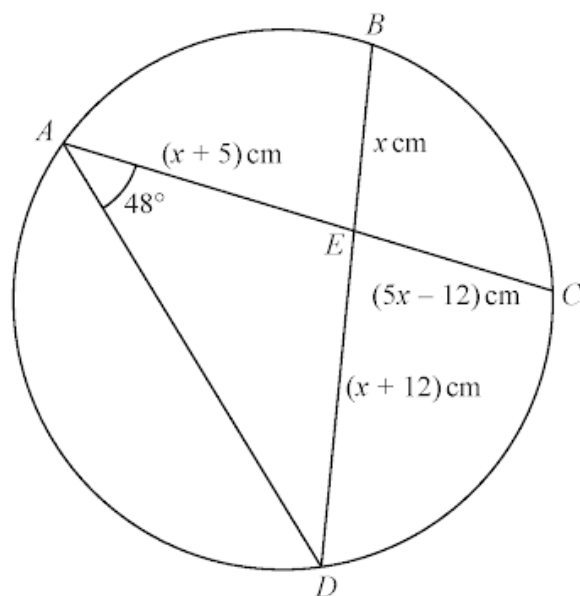


Diagram **NOT**
accurately drawn

$$AE = (x + 5) \text{ cm} \quad BE = x \text{ cm} \quad CE = (5x - 12) \text{ cm} \quad DE = (x + 12) \text{ cm}$$

$$\text{Angle } DAE = 48^\circ$$

Work out the size of angle ADE

Give your answer correct to one decimal place.

35. 4MA1/2HR_Winter_2023_Q18

A statue and a model of the statue are mathematically similar.

The statue has a total surface area of 3600 cm^2

The model has a total surface area of 625 cm^2

The volume of the model is 750 cm^3

Work out the volume of the statue.

..... cm^3

3 marks

A , B and C are points on a circle, centre O

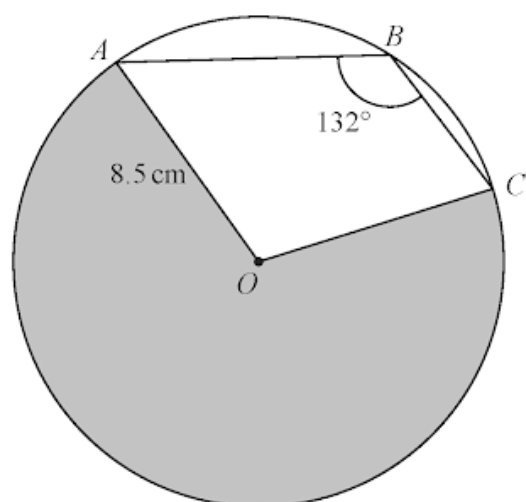


Diagram **NOT**
accurately drawn

The radius of the circle is 8.5 cm

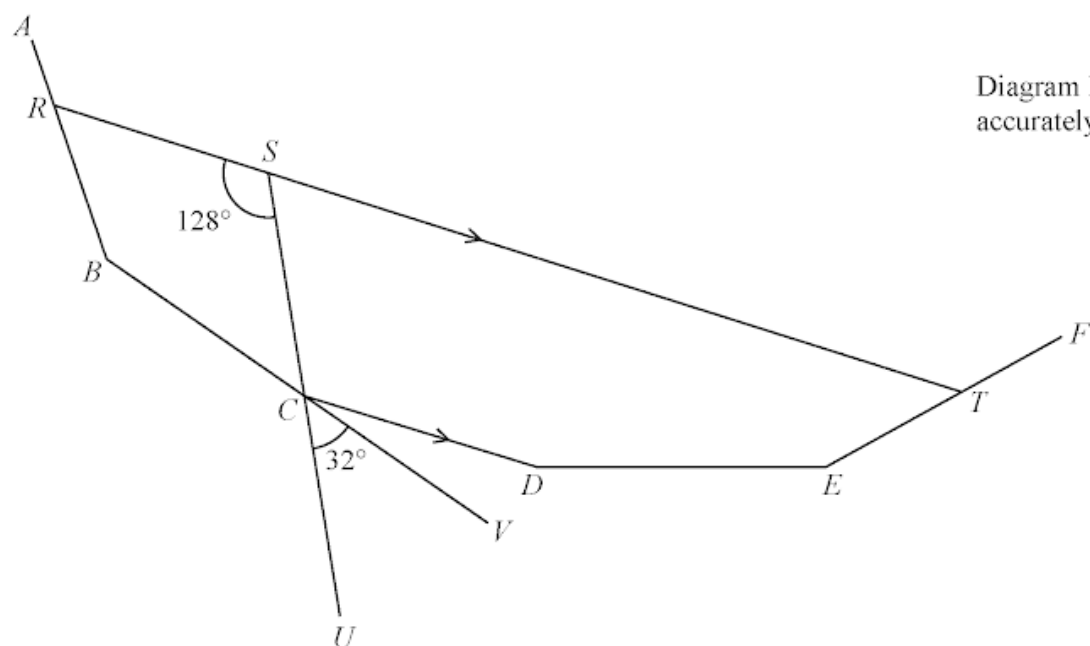
Angle $ABC = 132^\circ$

Work out the perimeter of the shaded sector AOC

Give your answer correct to 3 significant figures.

..... cm

3 marks



AB , BC , CD , DE and EF are five sides of a regular polygon.

RST , SCU and BCU are straight lines.

RST is parallel to CD

Angle $RSC = 128^\circ$

Angle $UCV = 32^\circ$

Work out how many sides the polygon has.

Show your working clearly.

4 marks

The diagram shows two solids, **A** and **B**, made from two different metals.

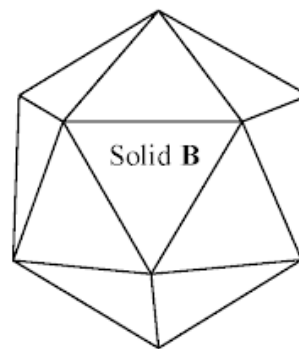
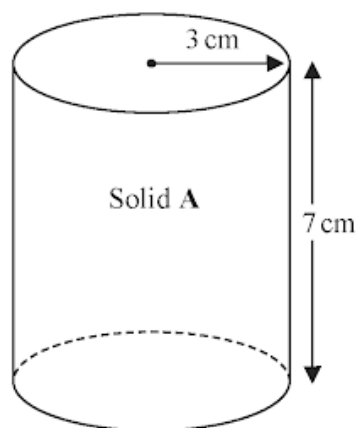


Diagram **NOT**
accurately drawn

Solid **A** is in the shape of a cylinder with radius 3 cm and height 7 cm

Solid **A** has a mass of 2000 g

Solid **B** has a mass of 3375 g

Solid **B** has a volume of 450 cm^3

All of the metal from Solid **A** and Solid **B** is melted down to make a uniform Solid **C**

Given that there is no change to mass or volume during this process

work out the density of Solid **C**

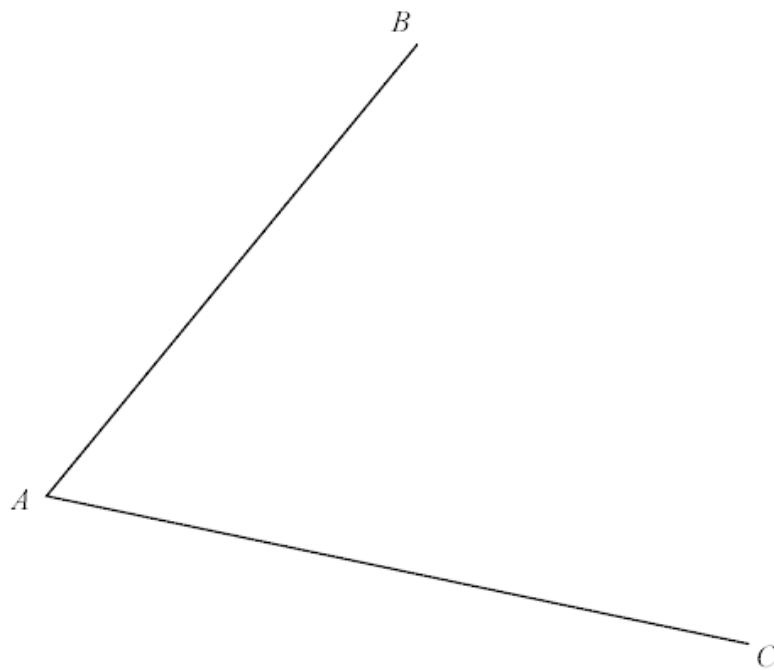
Give your answer correct to one decimal place.

..... g/cm³

3 marks

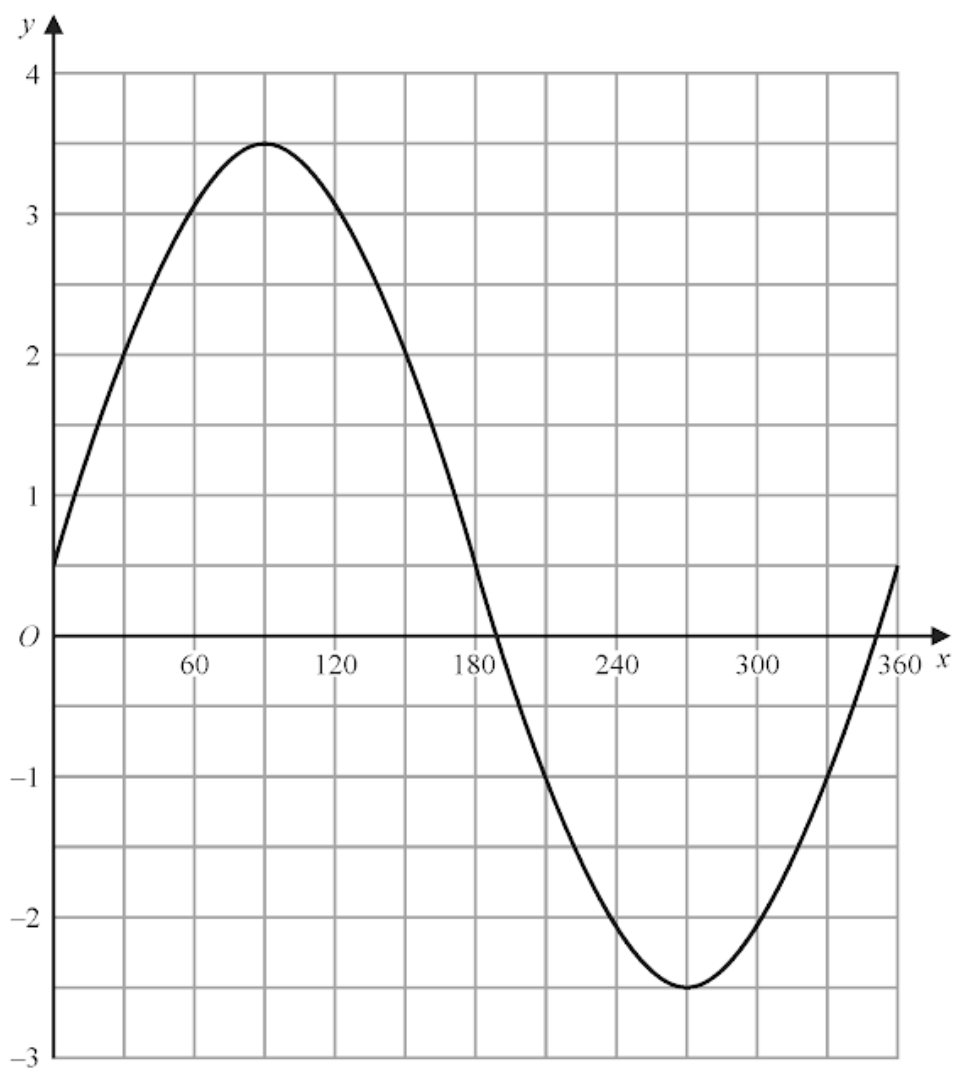
39. 4MA1/2HR_Winter_2023_Q4

Using ruler and compasses only, construct the bisector of angle BAC
You must show all your construction lines.



2 marks

The graph of $y = a \sin x^\circ + b$ is drawn on the grid.



Find the value of a and the value of b

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots$$

2 marks

The diagram shows a solid prism $ABCDEFGHJ$

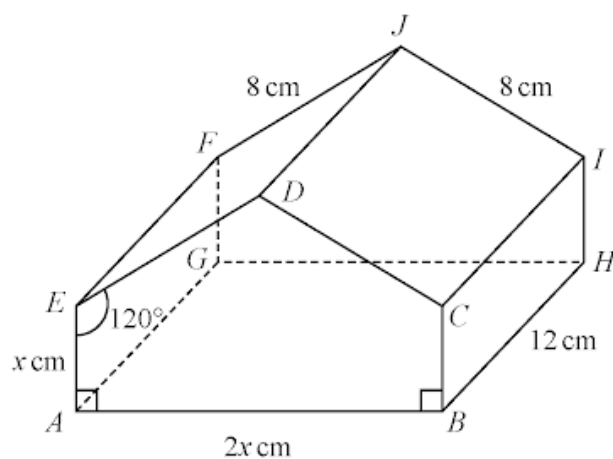


Diagram **NOT**
accurately drawn

The prism is such that each cross section is a pentagon where

$$AE = BC = x \text{ cm}$$

$$AB = 2x \text{ cm}$$

$$ED = CD = 8 \text{ cm}$$

$$\text{angle } EAB = \text{angle } CBA = 90^\circ$$

$$\text{angle } AED = \text{angle } BCD = 120^\circ$$

Given that $AG = BH = EF = DJ = CI = 12 \text{ cm}$

calculate the angle that AJ makes with the base $ABHG$ of the prism.

Give your answer correct to 3 significant figures.

42. 4MA1/2H_Winter_2023_Q21

Given that the surface area of a sphere is $49\pi \text{ cm}^2$

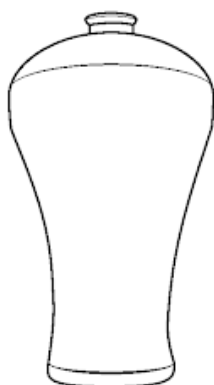
find the volume of the sphere.

Give your answer correct to the nearest integer.

..... cm^3

3 marks

A and **B** are two similar vases.



A



B

Diagram **NOT**
accurately drawn

The vases are such that

$$\text{surface area of vase B} = \frac{25}{64} \times \text{surface area of vase A}$$

and that

$$\text{volume of vase A} - \text{volume of vase B} = 541.8 \text{ cm}^3$$

Calculate the volume of vase **B**

..... cm^3

4 marks

A, B, C and D are points on a circle, centre O

EBF is the tangent to the circle at B

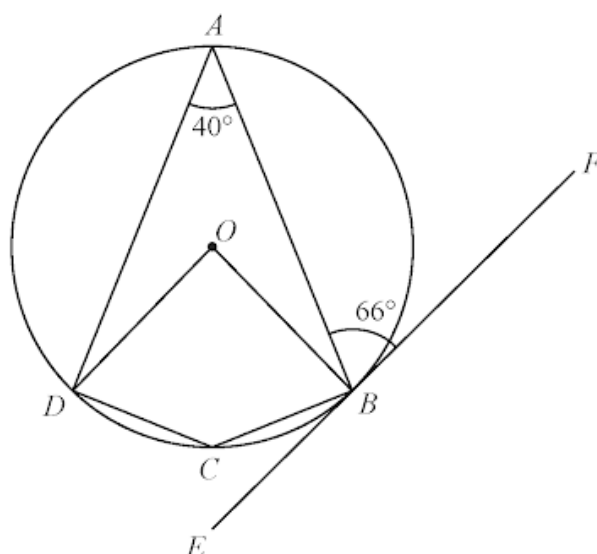


Diagram **NOT**
accurately drawn

(a) (i) Work out the size of angle DCB

(1)

(ii) Give a reason for your answer to (a)(i)

(1)

(b) Work out the size of angle ADO

(3)

The diagram shows an isosceles triangle, with base length 24 cm.

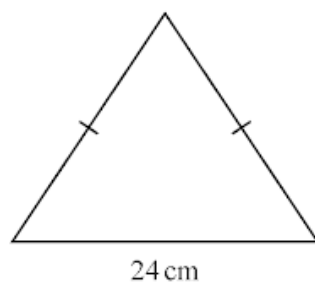


Diagram **NOT**
accurately drawn

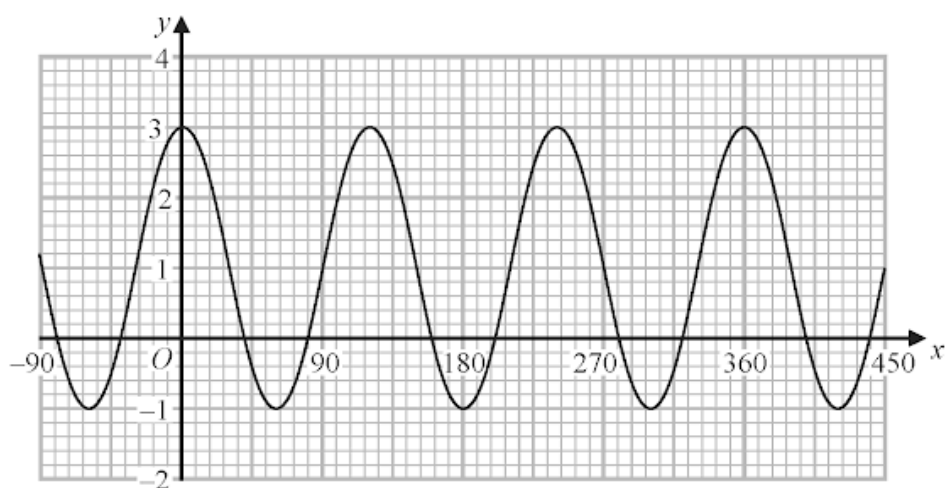
The perimeter of the triangle is 54 cm.

Work out the area of the triangle.

..... cm²

5 marks

Here is a sketch of the curve with equation $y = a \cos bx^\circ + c$ where $-90 \leq x \leq 450$



Find the value of a , the value of b and the value of c

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots$$

$$c = \dots\dots\dots$$

3 marks

47. 4MA1/2HR_Summer_2023_Q24

The surface area of sphere **A** is nine times the surface area of sphere **B**

The difference between the volume of sphere **A** and the volume of sphere **B** is $117\pi \text{ cm}^3$

Find the radius of the smaller sphere.

Show your working clearly.

..... cm

5 marks

Here is a cuboid $ABCDEFGH$

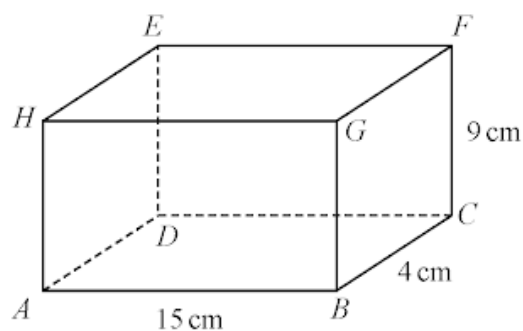


Diagram **NOT**
accurately drawn

$$AB = 15 \text{ cm} \quad BC = 4 \text{ cm} \quad CF = 9 \text{ cm}$$

- (a) Work out the length of BE
Give your answer correct to 3 significant figures.

..... cm

(2)

A , B and C are points on a circle.

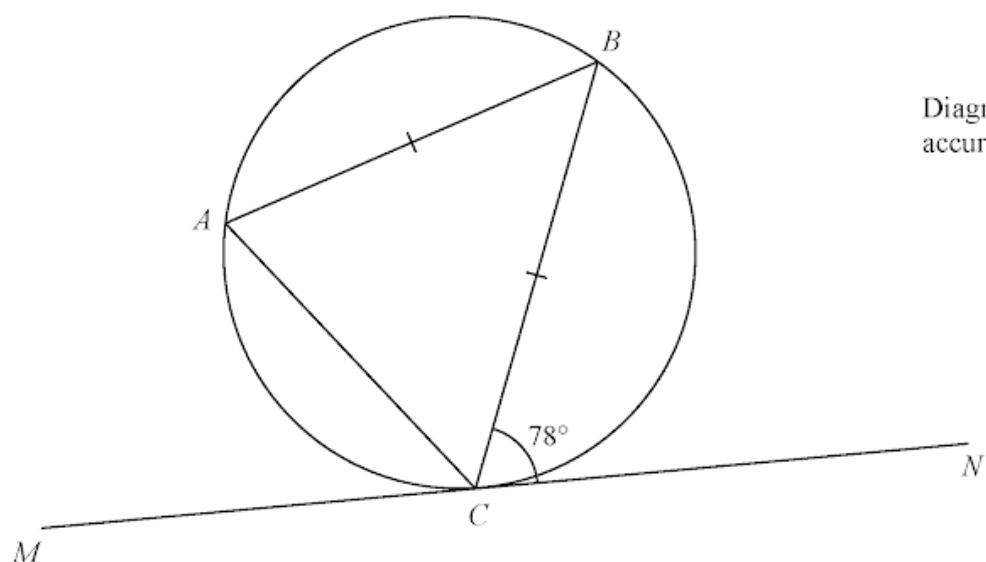


Diagram **NOT**
accurately drawn

MN is the tangent to the circle at C

$AB = CB$

Angle $BCN = 78^\circ$

Find the size of angle ABC

○

2 marks

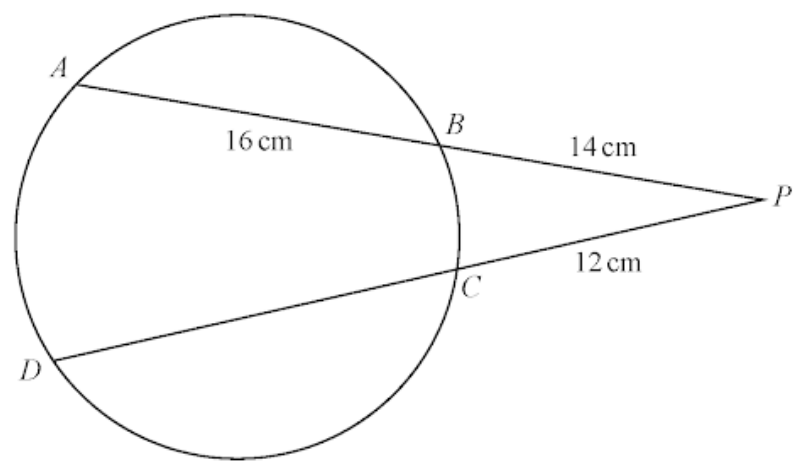


Diagram **NOT**
accurately drawn

A, B, C and D are points on a circle.

ABP and DCP are straight lines.

$AB = 16\text{ cm}$ $BP = 14\text{ cm}$ $CP = 12\text{ cm}$

Work out the length of DC

..... cm

3 marks

The diagram shows a rectangular sheet of metal $ABCD$

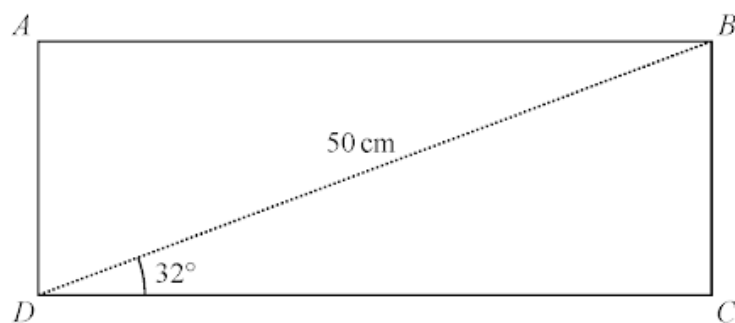


Diagram **NOT**
accurately drawn

$BD = 50\text{ cm}$ and angle $BDC = 32^\circ$

Nasser joins side AD to side BC to form a cylinder.

BC is the height of the cylinder.

DC is the circumference of the cross section of the cylinder.

Work out the volume, in cm^3 , of the cylinder.

Give your answer correct to 3 significant figures.

A field is in the shape of a trapezium.

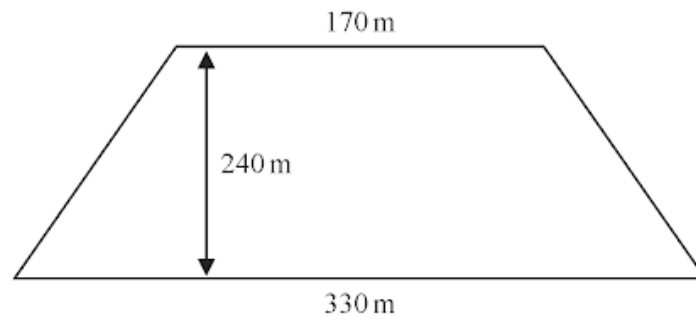


Diagram **NOT**
accurately drawn

The field is sold for a price of \$49 650

Given that 1 hectare = $10\,000\text{m}^2$

work out the average price of the field per hectare.

\$.....

4 marks

The diagram shows a cuboid with a square cross section.

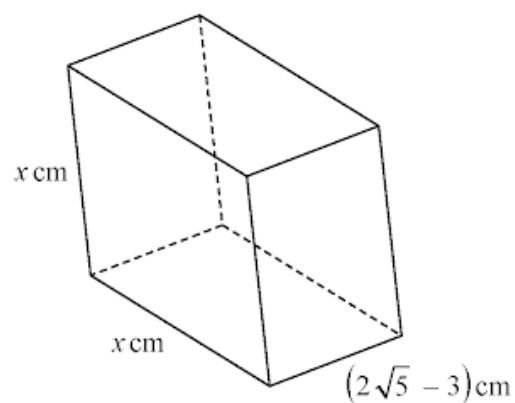


Diagram **NOT**
accurately drawn

The volume of the cuboid is $(13 + 6\sqrt{5})\text{cm}^3$

Without using a calculator, find the value of x

Give your answer in the form $a + \sqrt{b}$ where a and b are integers.

Show your working clearly.

The diagram shows a triangle ABC and a flagpole BF

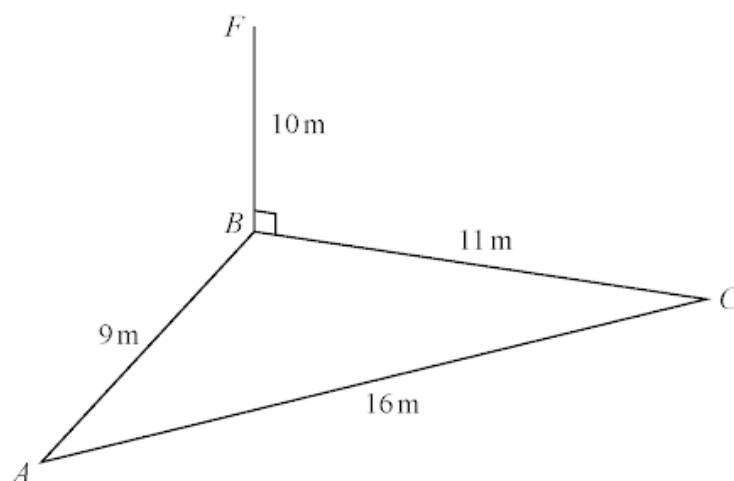


Diagram **NOT**
accurately drawn

A , B and C are points on horizontal ground.

BF is vertical.

$$AB = 9 \text{ m} \quad BC = 11 \text{ m} \quad AC = 16 \text{ m} \quad BF = 10 \text{ m}$$

D is the point on AC such that angle $BDC = 90^\circ$

Work out the size of the angle of elevation of the point F from the point D

Give your answer correct to one decimal place.

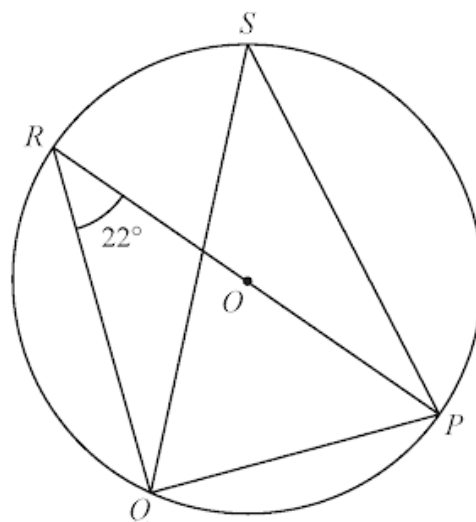


Diagram **NOT**
accurately drawn

P , Q , R and S are points on a circle, centre O
 ROP is a diameter of the circle.
 Angle $PRQ = 22^\circ$

(a) (i) Find the size of angle RQP

.....
 (1)

(ii) Give a reason for your answer.

.....

 (1)

(b) (i) Find the size of angle PSQ

.....
 (1)

(ii) Give a reason for your answer.

.....

 (1)

.....

The diagram shows right-angled triangle ABD

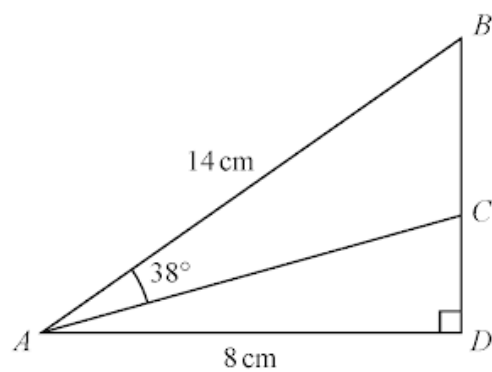


Diagram **NOT**
accurately drawn

$$AB = 14 \text{ cm} \quad AD = 8 \text{ cm}$$

C is the point on BD such that angle $BAC = 38^\circ$

Work out the length of CD

Give your answer correct to 3 significant figures.

..... cm

4 marks

Here is a 9-sided regular polygon $ABCDEFGHIJ$, with centre O

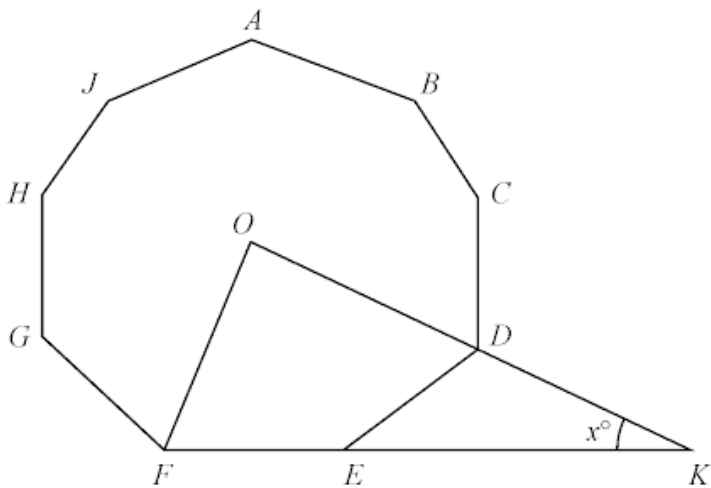


Diagram **NOT** accurately drawn

ODK and FEK are straight lines.
Work out the value of x

$x =$

3 marks

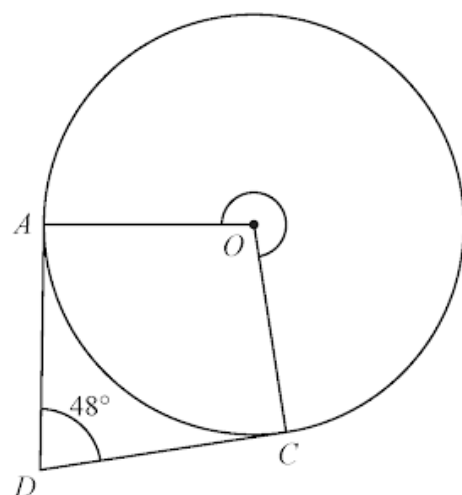


Diagram **NOT**
accurately drawn

A and C are points on a circle, centre O

DA is the tangent to the circle at A and DC is the tangent to the circle at C

Angle $ADC = 48^\circ$

Work out the size of reflex angle AOC

◇

3 marks

$ABCD$ is a trapezium.

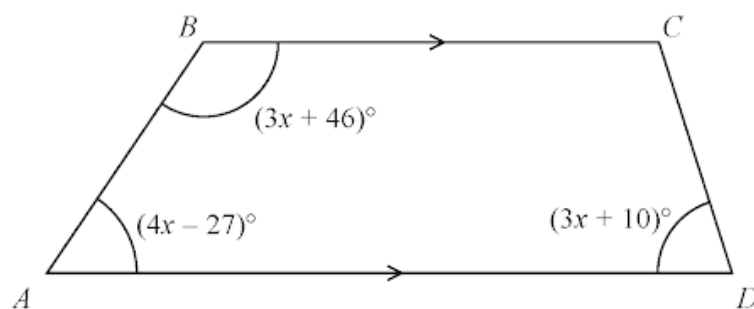


Diagram **NOT**
accurately drawn

BC is parallel to AD

Find the size of the largest angle inside the trapezium.

○

4 marks

ABC is a triangle in which angle $ABC = 90^\circ$

p and q are integers such that

the coordinates of A are $(p, 10)$

the coordinates of B are $(-1, -5)$

the coordinates of C are $(8, q)$

Given that the gradient of AC is $-\frac{6}{7}$

work out the value of p and the value of q

The diagram shows a solid made from a cylinder and a hemisphere.
The cylinder and the hemisphere are both made from the same metal.

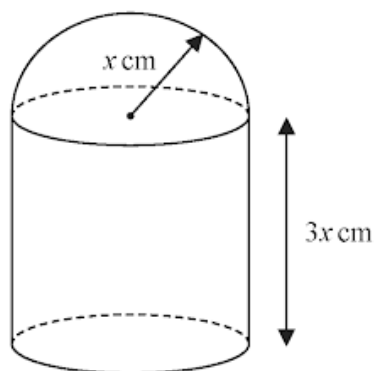


Diagram **NOT**
accurately drawn

The plane face of the hemisphere coincides with the upper plane face of the cylinder.

The radius of the cylinder and the radius of the hemisphere are both x cm.

The height of the cylinder is $3x$ cm.

The total surface area of the solid is $81\pi\text{ cm}^2$

The mass of the solid is 840 grams.

The following table gives the density of each of four metals.

Metal	Density (g/cm^3)
Aluminium	2.7
Nickel	8.9
Gold	19.3
Silver	10.5

The metal used to make the solid is one of the metals in the table.

Determine the metal used to make the solid.

Show your working clearly.

$ABCD$ is a horizontal rectangular field.

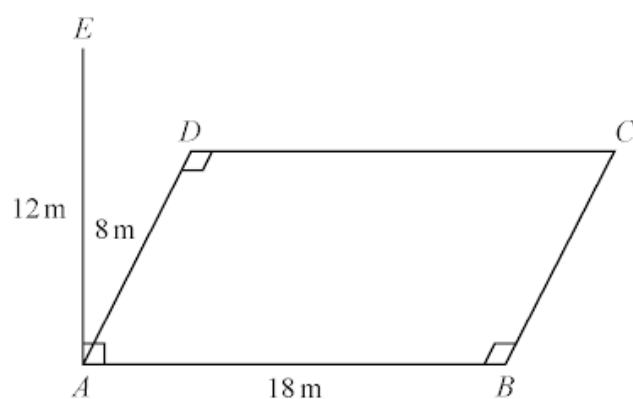


Diagram **NOT**
accurately drawn

A vertical pole, AE , is placed at the corner A of the field.

$$AE = 12\text{ m} \quad AB = 18\text{ m} \quad AD = 8\text{ m}$$

Calculate the size of the angle between EC and the plane $ABCD$.
Give your answer correct to one decimal place.

3 marks

The diagram shows quadrilateral $ABCD$

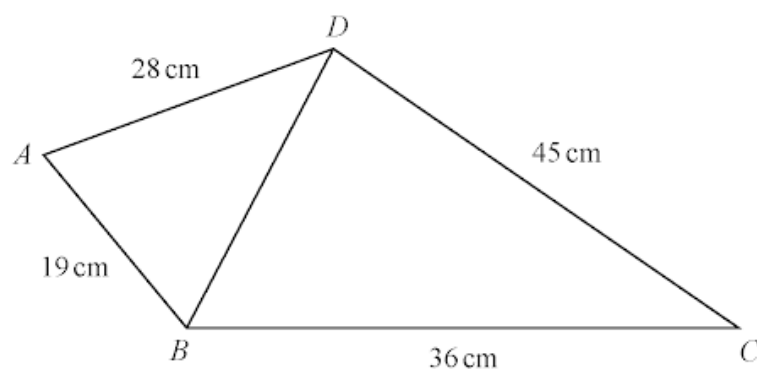


Diagram **NOT**
accurately drawn

The angle BCD is acute.

Given that the area of triangle $BCD = 405\text{ cm}^2$

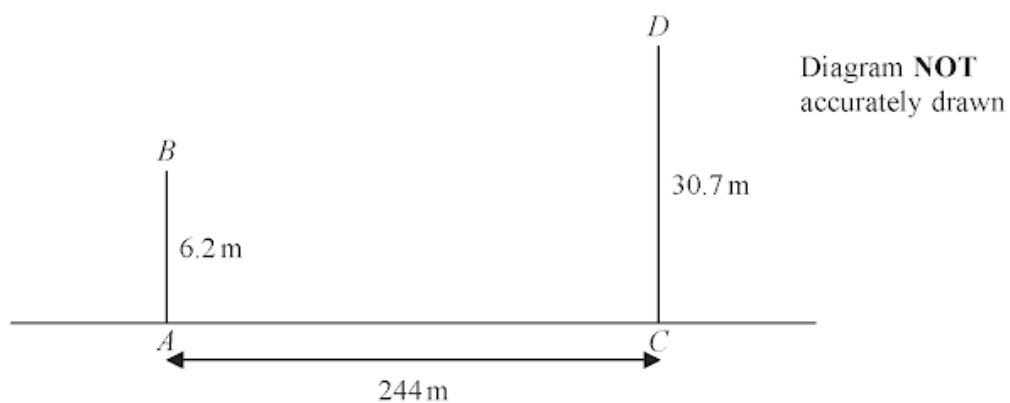
work out the size of angle ABD

Give your answer correct to one decimal place.

○

5 marks

The diagram shows two vertical phone masts, AB and CD , on horizontal ground.



$$AB = 6.2\text{ m} \quad AC = 244\text{ m} \quad CD = 30.7\text{ m}$$

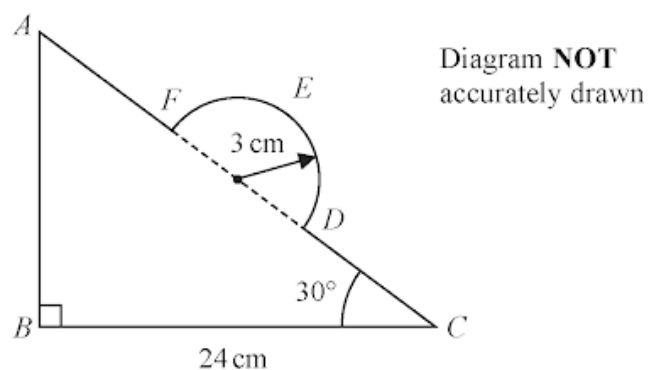
Work out the size of the angle of depression of B from D .

Give your answer correct to one decimal place.

○

3 marks

In the diagram, ABC is a right-angled triangle and DEF is a semicircular arc.



In triangle ABC

$$BC = 24 \text{ cm}$$

$$\text{angle } ABC = 90^\circ$$

$$\text{angle } BCA = 30^\circ$$

The points D and F lie on AC so that DF is the diameter of the semicircular arc DEF .
The radius of the semicircular arc is 3 cm.

Work out the length of $AFEDC$

Give your answer correct to 2 significant figures.

The diagram shows a regular octagon $ABCDEFGH$ and a regular pentagon $ABLIK$

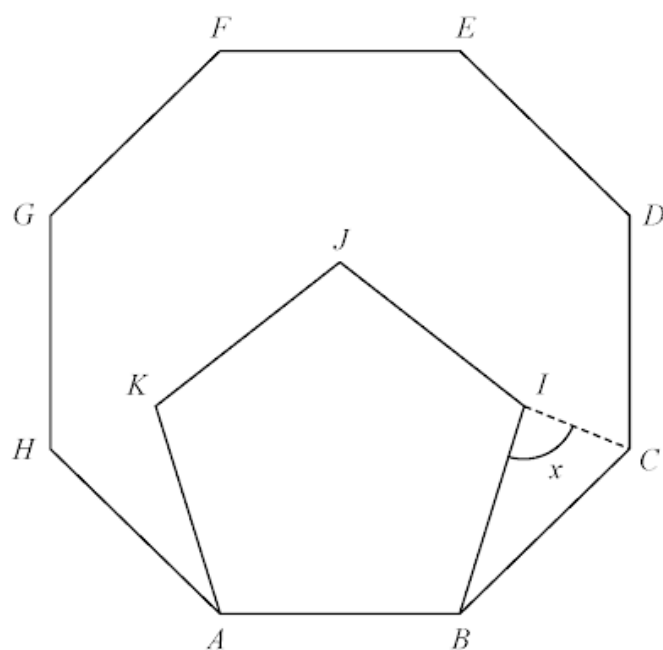


Diagram **NOT**
accurately drawn

Work out the size of the angle x

◊

4 marks

The diagram shows triangle PQR

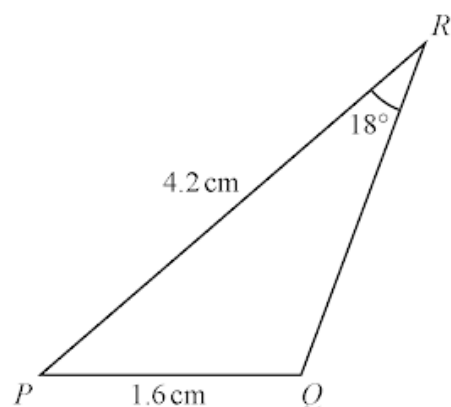


Diagram **NOT**
accurately drawn

$$PQ = 1.6 \text{ cm}$$

$$PR = 4.2 \text{ cm}$$

$$\text{Angle } PRQ = 18^\circ$$

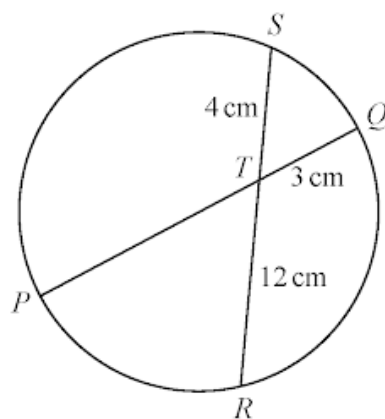
Given that angle PQR is obtuse,

work out the area of triangle PQR

Give your answer correct to 3 significant figures.

..... cm^2

6 marks

Diagram **NOT**
accurately drawn PTQ is a diameter of a circle. RTS is a chord of the circle.

$TQ = 3\text{ cm}$

$ST = 4\text{ cm}$

$TR = 12\text{ cm}$

Calculate the radius of the circle.

..... cm

3 marks

D, E, F and G are points on a circle, centre O

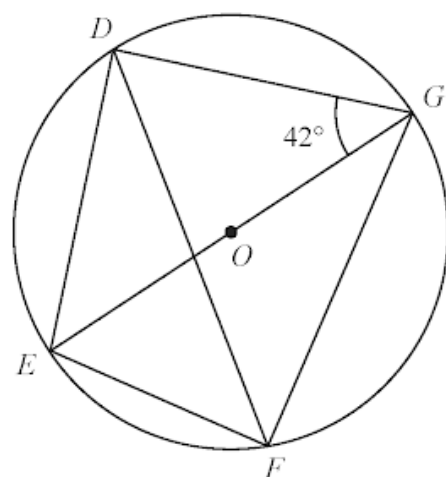


Diagram **NOT**
accurately drawn

EOG is a diameter of the circle.

Angle $EGD = 42^\circ$

Calculate the size of angle DFG

Give a reason for each stage of your working.

Angle $DFG = \dots\dots\dots^\circ$

4 marks

A zip wire is shown as the dashed line AC in the diagram.

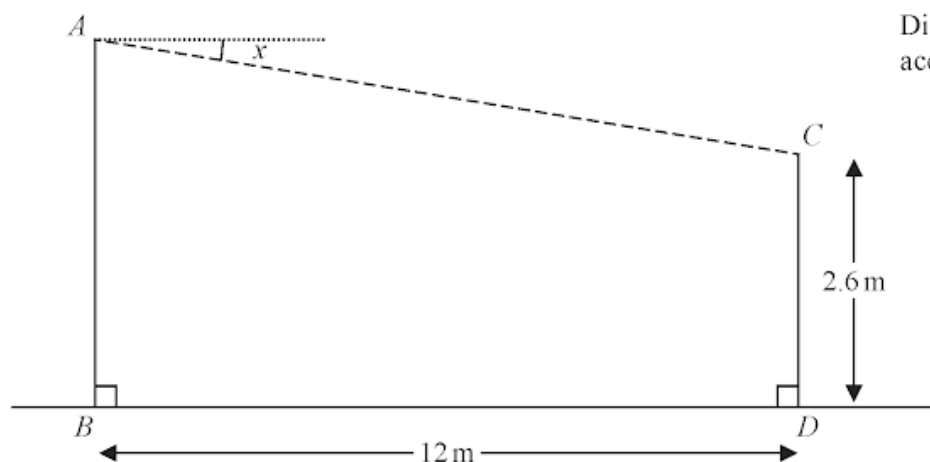


Diagram **NOT**
accurately drawn

The zip wire is supported by two vertical posts AB and CD standing on horizontal ground.

$$CD = 2.6\text{ m} \quad BD = 12\text{ m}$$

The zip wire makes an angle x with the horizontal, as shown in the diagram.
The design of the zip wire requires the angle x to be at least 5° .

Work out the least possible height of the post AB
Give your answer correct to 3 significant figures.

..... m

(Total for Question 14 is 3 marks)

The diagram shows a regular 10-sided polygon, $ABCDEFGHIJ$

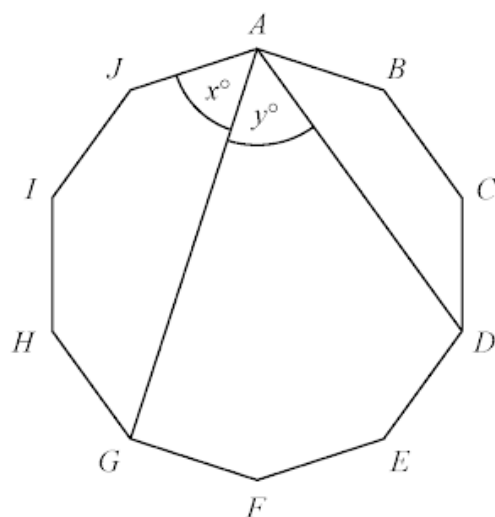
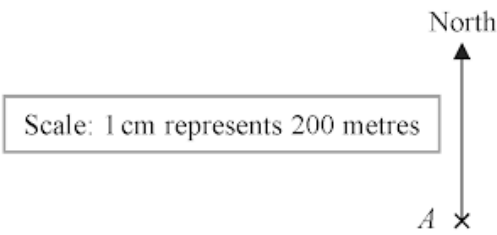


Diagram **NOT**
accurately drawn

Show that $x = y$

4 marks

The scale diagram shows the position on a map of a house, A



House C is on a bearing of 110° from A
The distance from A to C is 700 m

- (a) Mark the position of C on the diagram with a cross (×)
Label your cross C

(3)

- (b) Write the scale of the map in the form $1:n$

1 :
(1)

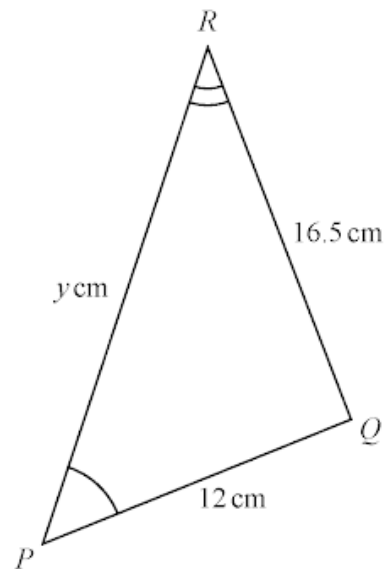
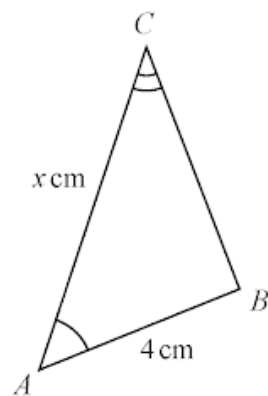


Diagram **NOT**
accurately drawn

Triangle ABC is similar to triangle PQR

$$AB = 4 \text{ cm} \quad PQ = 12 \text{ cm} \quad RQ = 16.5 \text{ cm} \quad AC = x \text{ cm} \quad PR = y \text{ cm}$$

(a) Calculate the length of BC

..... cm
(2)

(b) Write down an expression for y in terms of x

$y =$
(1)

The diagram shows a frustum of a cone, and a sphere.

The frustum, shown shaded in the diagram, is made by removing the small cone from the large cone.

The small cone and the large cone are similar.

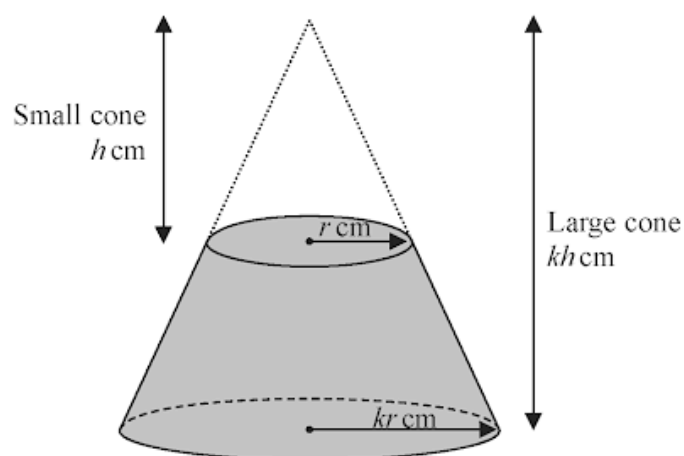


Diagram **NOT**
accurately drawn

The height of the small cone is h cm and the radius of the base of the small cone is r cm.
The height of the large cone is kh cm and the radius of the base of the large cone is kr cm.
The radius of the sphere is r cm.

The sphere is divided into two hemispheres, each of radius r cm.

Solid **A** is formed by joining one of the hemispheres to the frustum.

The plane face of the hemisphere coincides with the upper plane face of the frustum, as shown in the diagram below.

Solid **B** is formed by joining the other hemisphere to the small cone that was removed from the large cone.

The plane face of the hemisphere coincides with the plane face of the base of the small cone, as shown in the diagram below.

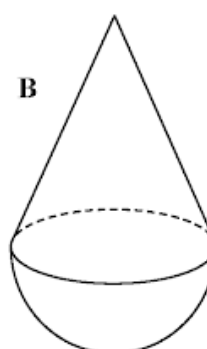
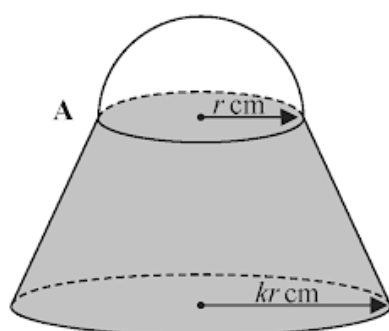


Diagram **NOT**
accurately drawn

The volume of solid **A** is 6 times the volume of solid **B**.

Given that $k > \sqrt[3]{7}$

find an expression for h in terms of k and r

$h = \dots\dots\dots$

6 marks

75. 4MA1/2HR_Summer_2022_Q20

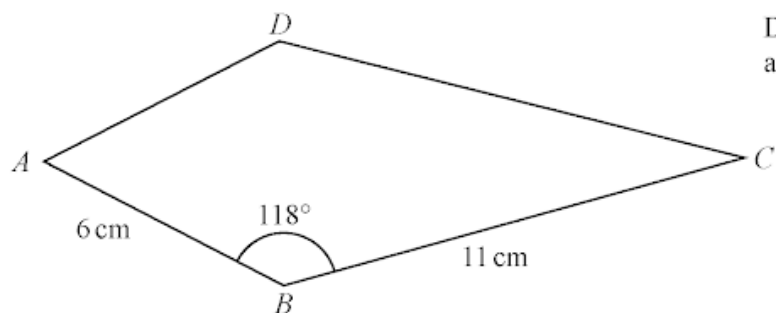
The centre O of a circle has coordinates $(4, 7)$

The point A , on the circle, has coordinates $(6, 11)$ and AOP is a diameter of the circle.

Find an equation of the tangent to the circle at the point P

.....
4 marks

The diagram shows a kite $ABCD$



$$AB = 6 \text{ cm}$$

$$BC = 11 \text{ cm}$$

$$\text{Angle } ABC = 118^\circ$$

Calculate the area of the kite.

Give your answer correct to 3 significant figures.

..... cm^2

3 marks

The diagram shows a sector AOB of a circle with centre O

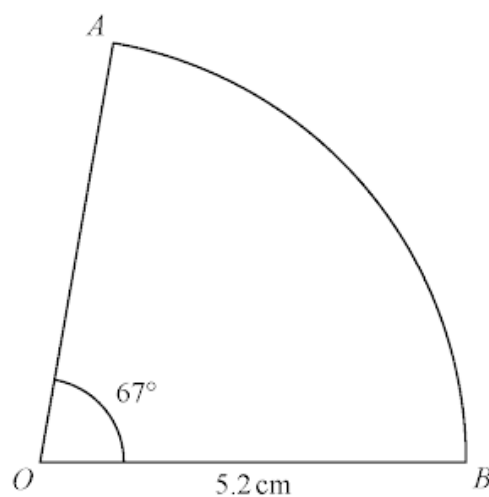


Diagram **NOT**
accurately drawn

Angle $AOB = 67^\circ$
 $OA = OB = 5.2$ cm

Calculate the perimeter of the sector.
Give your answer correct to 3 significant figures.

..... cm

3 marks

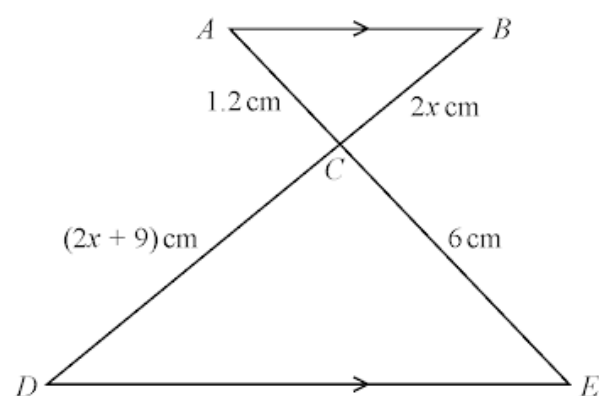


Diagram **NOT**
accurately drawn

ACE and BCD are straight lines.
 AB is parallel to DE

Work out the value of x

$x = \dots\dots\dots$

3 marks

The diagram shows triangle ABP inside the regular hexagon $ABCDEF$

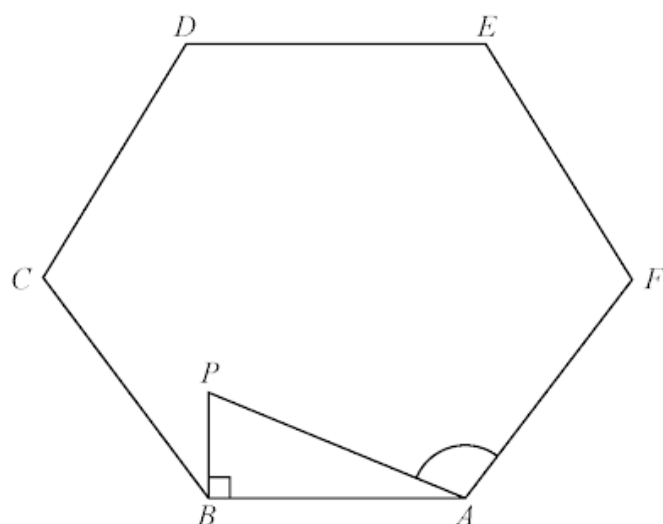


Diagram **NOT**
accurately drawn

$$AB = 5 \text{ cm}$$

$$BP = 2 \text{ cm}$$

$$\text{Angle } ABP = 90^\circ$$

Work out the size of angle PAF

Give your answer correct to 3 significant figures.

○

5 marks

The diagram shows a shape made from a square $ABCD$ and 4 identical semicircles.

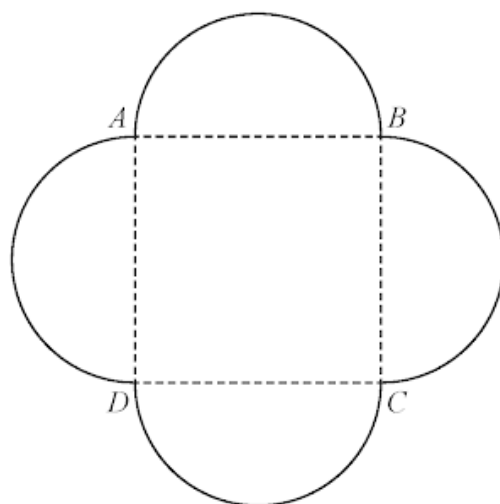


Diagram **NOT**
accurately drawn

As shown in the diagram, the semicircles have AB , BC , CD and DA as diameters.

The area of the square is 36 cm^2

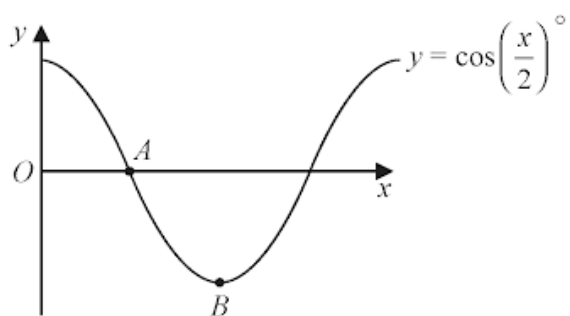
Calculate the total area of the shape.

Give your answer correct to one decimal place.

..... cm^2

4 marks

The diagram shows a sketch of the graph of $y = \cos\left(\frac{x}{2}\right)^\circ$



(i) Find the coordinates of the point A

(.....,)
(1)

(ii) Find the coordinates of the point B

(.....,)
(1)

The diagram shows four identical circles drawn inside a square.

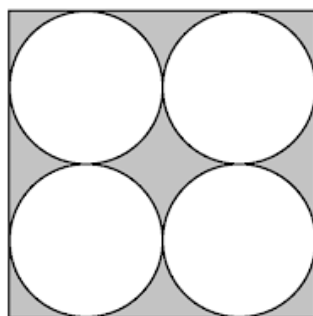


Diagram **NOT**
accurately drawn

Each circle touches two other circles and two sides of the square.

The region inside the square that is outside the circles, shown shaded in the diagram, has a total area of 40 cm^2

Work out the perimeter of the square.

Give your answer correct to 3 significant figures.

..... cm

4 marks

The diagram shows triangle PQR .

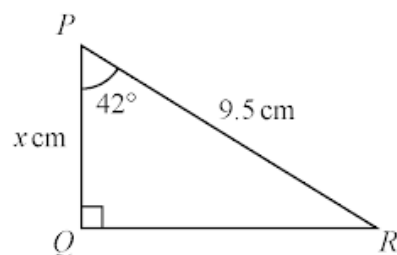


Diagram **NOT**
accurately drawn

Work out the value of x

Give your answer correct to one decimal place.

$x =$

3 marks

The diagram shows an 8-sided shape $ABCDEFGH$.

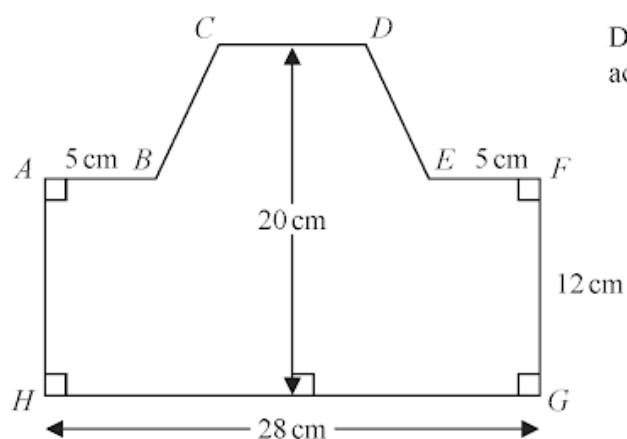


Diagram **NOT**
accurately drawn

$$HG = 28 \text{ cm} \quad FG = 12 \text{ cm} \quad AB = EF = 5 \text{ cm}$$

The height of the shape is 20 cm

CD is parallel to HG

The area of shape $ABCDEFGH$ is 434 cm^2

Find the length of CD .

..... cm

4 marks

85. 4MA1/2HR_Winter_2021_Q22

ABC is an isosceles triangle in a horizontal plane.
The point T is vertically above B .

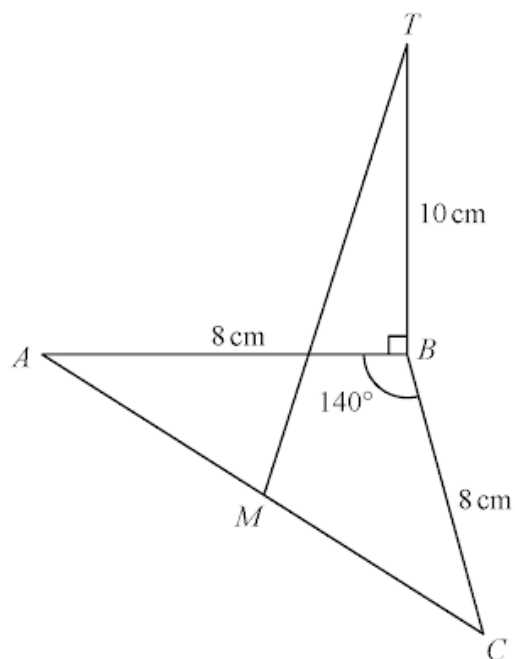


Diagram **NOT**
accurately drawn

Angle $ABC = 140^\circ$

$AB = BC = 8\text{ cm}$

$TB = 10\text{ cm}$

M is the midpoint of AC .

Calculate the size of the angle between MT and the horizontal plane ABC .

Give your answer correct to one decimal place.

A , B and C are points on a circle with centre O .

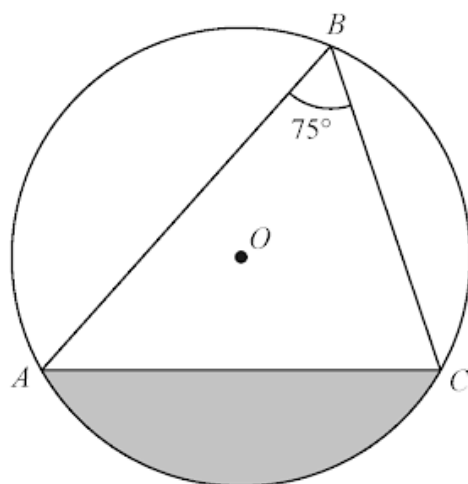


Diagram **NOT**
accurately drawn

Angle $ABC = 75^\circ$

The area of the shaded segment is 200 cm^2

Calculate the radius of the circle.

Give your answer correct to 3 significant figures.

..... cm

(Total for Question is 5 marks)

The diagram shows two hot air balloons.

A is a point on the base of one of the balloons and B is a point on the base of the other balloon.

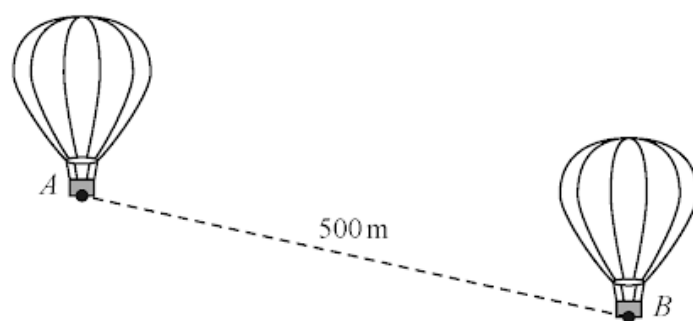


Diagram **NOT**
accurately drawn

The distance between A and B is 500 metres.

The angle of depression of B from A is 23° .

Calculate the vertical height of A above B .

Give your answer correct to one decimal place.

..... metres

(Total for Question is 3 marks)

A circle centre O has radius 9 cm.

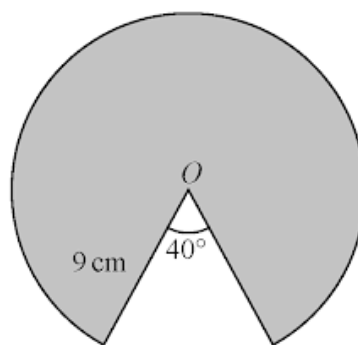


Diagram **NOT**
accurately drawn

Calculate the perimeter of the shaded sector of the circle.
Give your answer correct to 3 significant figures.

..... cm

(Total for Question is 4 marks)

The diagram shows a rectangle $ABCD$ and a semicircle with diameter AB where $AB = 12$ cm. The point E lies on DC and also on the semicircle.

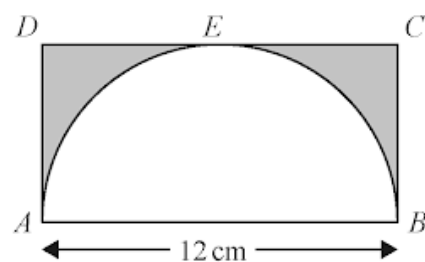


Diagram **NOT**
accurately drawn

Work out the area of the shaded region.
Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question is 3 marks)

The diagram shows a sector OBC of a circle with centre O and radius $(6 + x)$ cm.

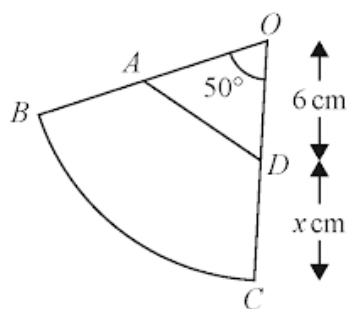


Diagram **NOT**
accurately drawn

A is the point on OB and D is the point on OC such that $OA = OD = 6$ cm

Angle $BOC = 50^\circ$

Given that

the perimeter of sector $OBC = 2 \times$ the perimeter of triangle OAD

find the value of x .

Give your answer correct to 3 significant figures.

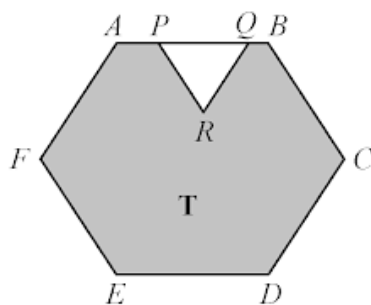


Diagram **NOT**
accurately drawn

The diagram shows a shaded region **T** formed by removing an equilateral triangle PQR from a regular hexagon $ABCDEF$.

The points P and Q lie on AB such that $AB = 1.5 \times PQ$

Given that the area of region **T** is $72\sqrt{3} \text{ cm}^2$

work out the length of PQ .

..... cm

(Total for Question is 4 marks)

The diagram shows a solid prism $ABCDEFGH$.

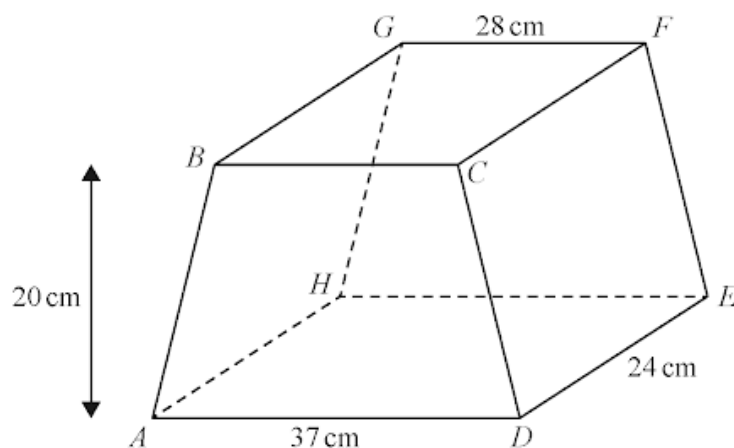


Diagram **NOT**
accurately drawn

The trapezium $ABCD$, in which AD is parallel to BC , is a cross section of the prism.

The base $ADEH$ of the prism is a horizontal plane.

$ADEH$ and $BCFG$ are rectangles.

The midpoint of BC is vertically above the midpoint of AD so that $BA = CD$.

$$AD = 37 \text{ cm} \qquad GF = 28 \text{ cm} \qquad DE = 24 \text{ cm}$$

The perpendicular distance between edges AD and BC is 20 cm.

(a) Work out the total surface area of the prism.

..... cm^2
(4)

93. 4MA1/2H_Winter_2021_Q13

P, Q, R, S and T are points on a circle with centre O .

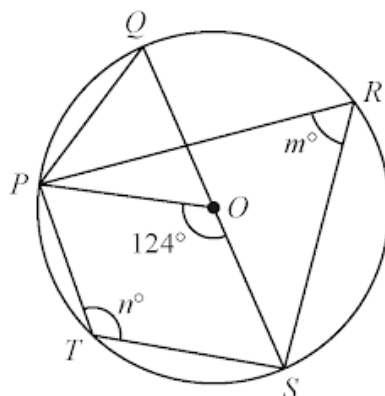


Diagram **NOT**
accurately drawn

QOS is a diameter of the circle.

angle $POS = 124^\circ$ angle $PRS = m^\circ$ angle $PTS = n^\circ$

(a) Find the value of

(i) m

(ii) n

(b) Find the size of angle QPO .

(2)

(1)

A , B and C are points on a circle with centre O .

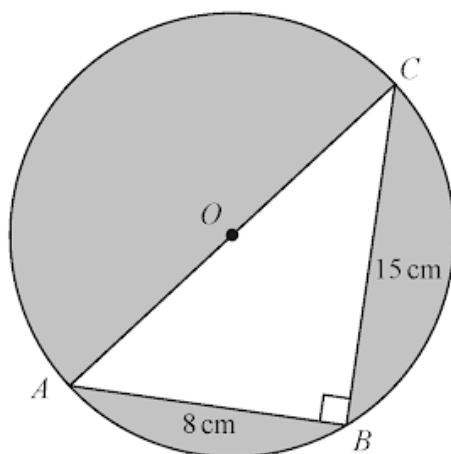


Diagram **NOT**
accurately drawn

AOC is a diameter of the circle.

$AB = 8\text{ cm}$ $BC = 15\text{ cm}$

Angle $ABC = 90^\circ$

Work out the total area of the regions shown shaded in the diagram.
Give your answer correct to 3 significant figures.

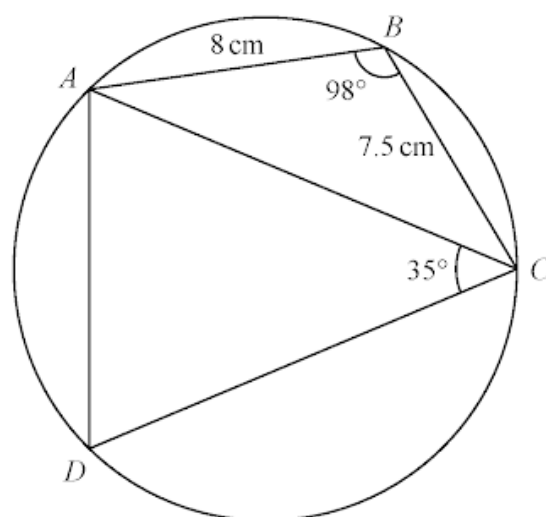


Diagram **NOT**
accurately drawn

$ABCD$ is a quadrilateral where A , B , C and D are points on a circle.

$$AB = 8 \text{ cm}$$

$$BC = 7.5 \text{ cm}$$

$$\text{Angle } ABC = 98^\circ$$

$$\text{Angle } ACD = 35^\circ$$

Work out the perimeter of quadrilateral $ABCD$.

Give your answer correct to one decimal place.

The straight line **L** passes through the points $(4, -1)$ and $(6, 4)$

The straight line **M** is perpendicular to **L** and intersects the y -axis at the point $(0, 8)$

Find the coordinates of the point where **M** intersects the x -axis.

(.....,)

4 marks

Here is a triangular prism.

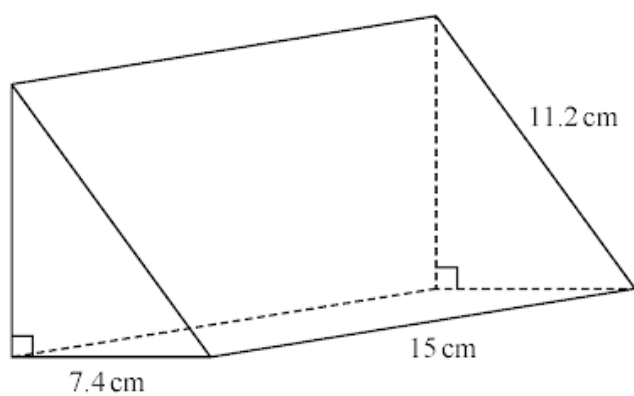


Diagram **NOT**
accurately drawn

Work out the volume of the prism.
Give your answer correct to 3 significant figures.

..... cm^3

5 marks

Here is a right-angled triangle.

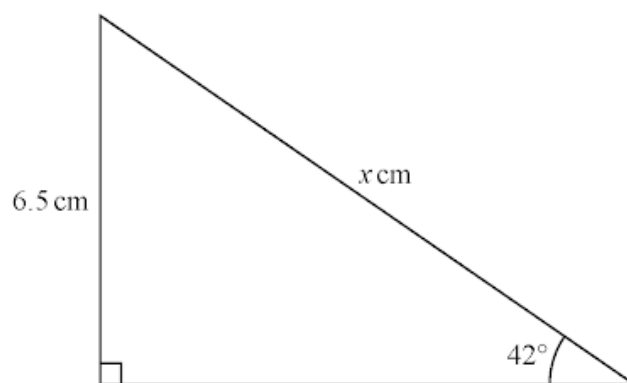


Diagram **NOT**
accurately drawn

Work out the value of x .
Give your answer correct to one decimal place.

$x =$

3 marks

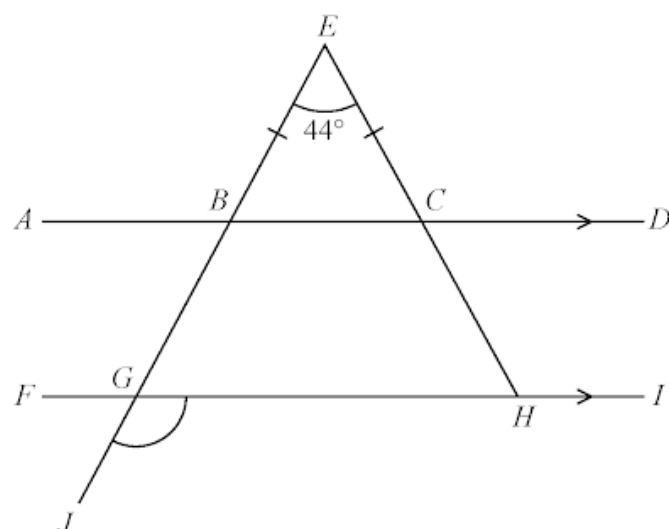


Diagram **NOT**
accurately drawn

$ABCD$ and $FGHI$ are parallel straight lines.
 $EBGJ$ and ECH are straight lines.

$BE = CE$
 Angle $BEC = 44^\circ$

Work out the size of angle JGH .
 Give a reason for each stage of your working.

○

5 marks

100. 4MA1/2H_Summer_2021_Q3

The bearing of Paris from London is 149°

Work out the bearing of London from Paris.



2 marks

2020

The diagram shows two similar vases, **A** and **B**.

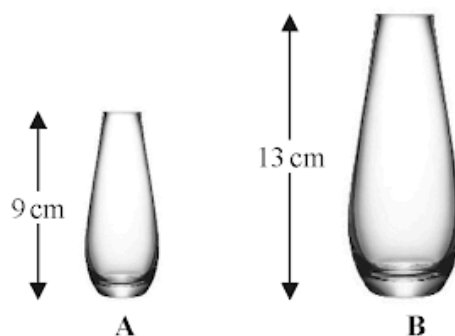


Diagram **NOT**
accurately drawn

The height of vase **A** is 9 cm and the height of vase **B** is 13 cm.

Given that

$$\text{surface area of vase A} + \text{surface area of vase B} = 1800 \text{ cm}^2$$

calculate the surface area of vase **A**.

..... cm^2

(Total for Question is 4 marks)

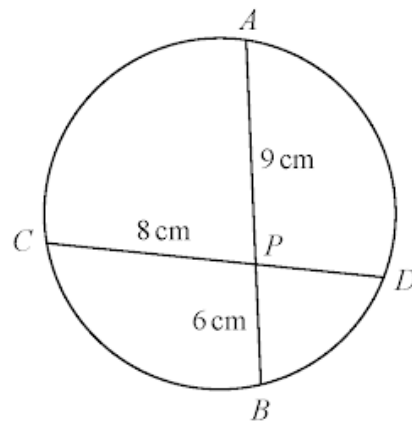


Diagram **NOT**
accurately drawn

APB and CPD are chords of a circle.

$AP = 9\text{ cm}$ $PB = 6\text{ cm}$ $CP = 8\text{ cm}$

Calculate the length of PD .

..... cm

(Total for Question is 2 marks)

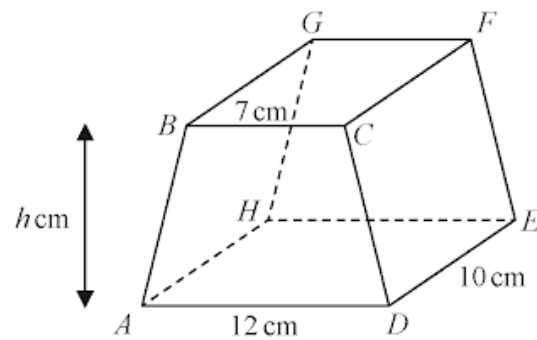


Diagram **NOT**
accurately drawn

The diagram shows a prism $ABCDEFGH$ in which $ABCD$ is a trapezium with BC parallel to AD and $CDEF$ is a rectangle.

$$BC = 7 \text{ cm} \quad AD = 12 \text{ cm} \quad DE = 10 \text{ cm}$$

The height of trapezium $ABCD$ is $h \text{ cm}$

The volume of the prism is 608 cm^3

Work out the value of h .

$$h = \dots\dots\dots$$

(Total for Question is 3 marks)

The diagram shows two congruent isosceles triangles and parts of two congruent regular polygons, **X** and **Y**.

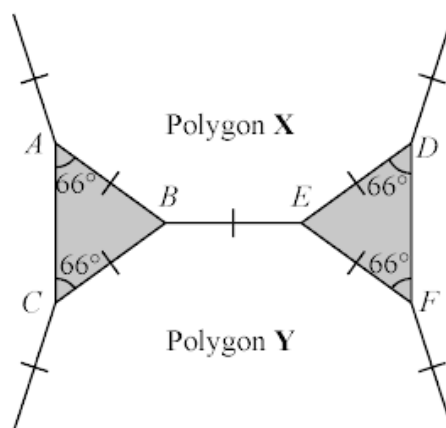


Diagram **NOT**
accurately drawn

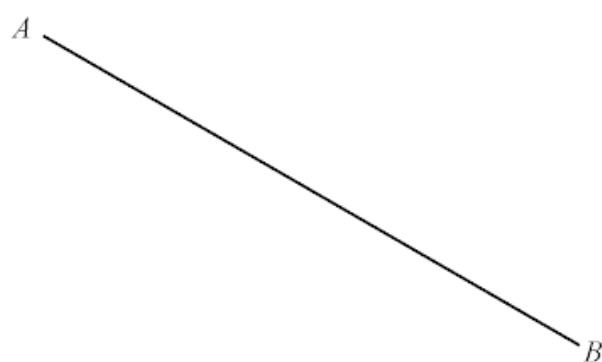
The two regular polygons each have n sides.

Work out the value of n .

$$n = \dots\dots\dots$$

(Total for Question is 3 marks)

Use ruler and compasses only to construct the perpendicular bisector of the line AB .
You must show all your construction lines.



(Total for Question is 2 marks)

The diagram shows a rectangle and a diagonal of the rectangle.

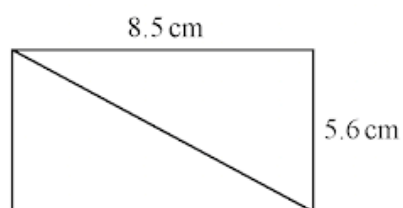


Diagram **NOT**
accurately drawn

Work out the length of the diagonal of the rectangle.
Give your answer correct to 1 decimal place.

..... cm

(Total for Question is 3 marks)

Here is a sector, AOB , of a circle with centre O and angle $AOB = x^\circ$

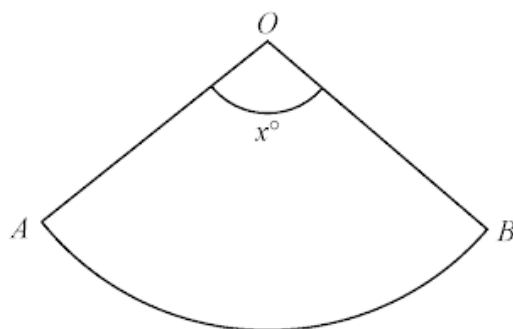


Diagram **NOT**
accurately drawn

The sector can form the curved surface of a cone by joining OA to OB .

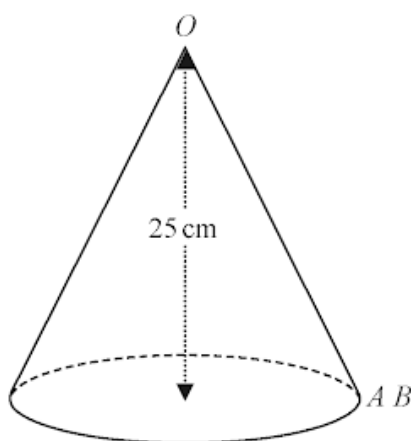


Diagram **NOT**
accurately drawn

The height of the cone is 25 cm .

The volume of the cone is 1600 cm^3

Work out the value of x .

Give your answer correct to the nearest whole number.

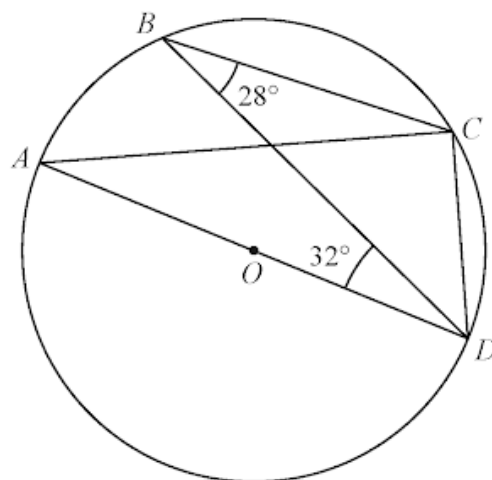


Diagram **NOT**
accurately drawn

A , B , C and D are points on a circle, centre O .
 AOD is a diameter of the circle.

Angle $CBD = 28^\circ$

Angle $BDA = 32^\circ$

Find the size of angle BDC .

Give a reason for each stage of your working.

○

(Total for Question 13 is 4 marks)

The diagram shows a shaded shape $ABCD$ made from a semicircle ABC and a right-angled triangle ACD .

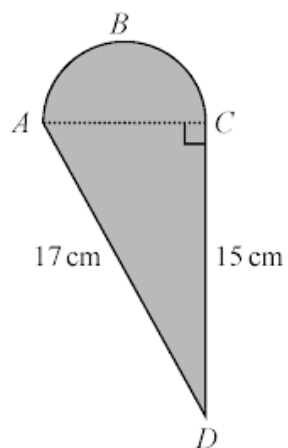


Diagram **NOT**
accurately drawn

AC is the diameter of the semicircle ABC .

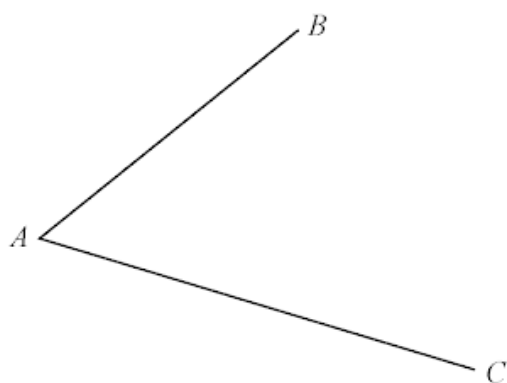
Work out the perimeter of the shaded shape.

Give your answer correct to 3 significant figures.

..... cm

(Total for Question is 5 marks)

Use ruler and compasses to construct the bisector of angle BAC .
You must show all your construction lines.



(Total for Question is 2 marks)

The diagram shows a triangle.

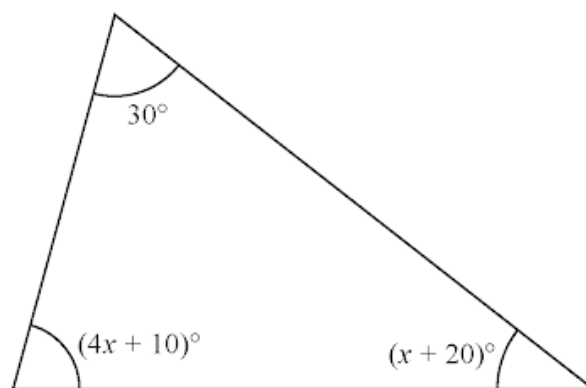


Diagram **NOT**
accurately drawn

Work out the value of x .

$x =$

(Total for Question is 4 marks)

A, B, D and E are points on a circle.
 ABC and EDC are straight lines.

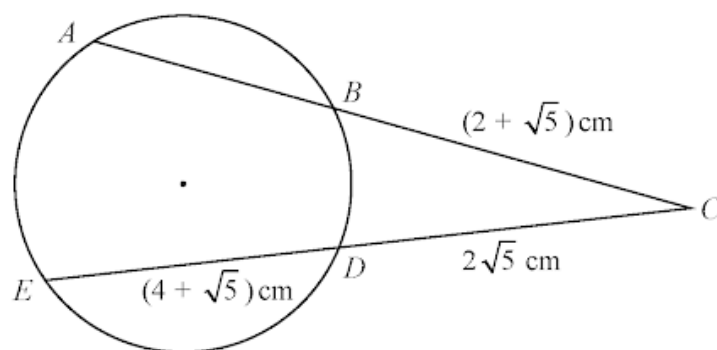


Diagram **NOT**
 accurately drawn

$$BC = (2 + \sqrt{5}) \text{ cm}$$

$$ED = (4 + \sqrt{5}) \text{ cm}$$

$$DC = 2\sqrt{5} \text{ cm}$$

Show that the length of AB is $(p\sqrt{5} + q) \text{ cm}$, where p and q are integers whose values are to be found.

Show your working clearly.

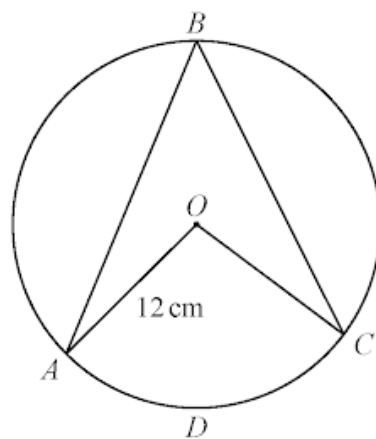


Diagram **NOT**
accurately drawn

A , B , C and D are points on a circle with centre O and radius 12 cm.

The area of the sector $OADC$ of the circle is 100 cm^2

Work out the size of angle ABC .

Give your answer correct to 3 significant figures.

(Total for Question is 4 marks)

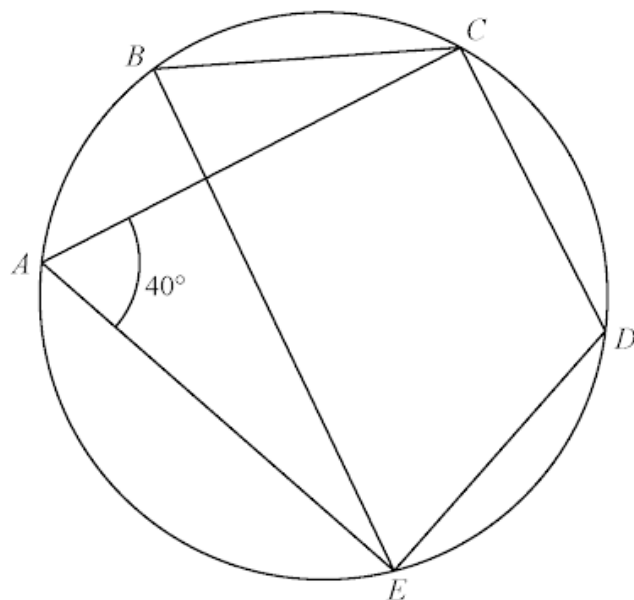


Diagram **NOT**
accurately drawn

A, B, C, D and E are points on a circle.

Angle $EAC = 40^\circ$

(a) (i) Write down the size of angle EBC .

.....
(1)

(ii) Give a reason for your answer.

.....
(1)

(b) Find the size of angle EDC .

.....
(1)

Platinum nuggets are in the shape of a solid cylinder.

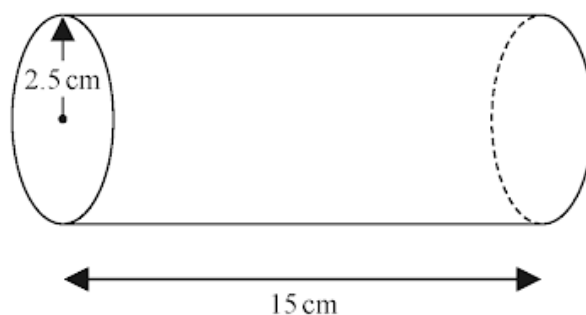


Diagram **NOT**
accurately drawn

The radius of each cylinder is 2.5 cm.

The length of each cylinder is 15 cm.

The density of platinum is 21.5 g/cm^3

The greatest mass that Jacques can carry is 30 kg.

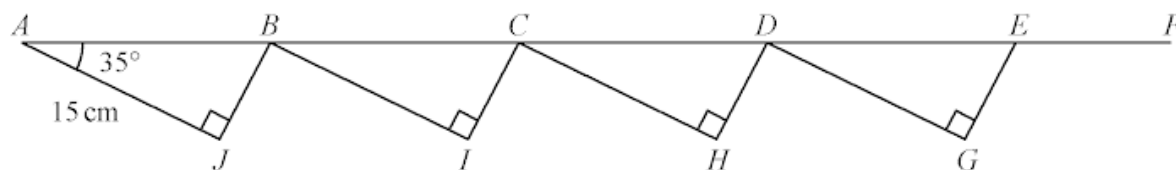
Can Jacques carry 5 platinum nuggets at the same time?

You must show all your working.

(Total for Question is 5 marks)

The diagram shows four congruent right-angled triangles ABJ , BCI , CDH and DEG .
The diagram also shows the straight line $ABCDEF$.

Diagram **NOT**
accurately drawn



$$AJ = 15 \text{ cm}$$

$$\text{Angle } BAJ = 35^\circ$$

$$AF = 80 \text{ cm}$$

Work out the length of EF .

Give your answer correct to 3 significant figures.

..... cm

(Total for Question is 5 marks)

Markus makes a steel framework.

The framework is in the shape of the right-angled triangle ABC shown in the diagram.

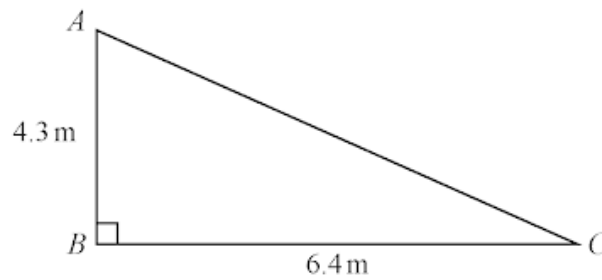


Diagram **NOT**
accurately drawn

The steel that Markus uses costs \$22 per metre.

The steel can **only** be bought in a length that is a whole number of metres.

Work out the total cost of the steel that Markus buys in order to make the framework.

\$.....

(Total for Question is 4 marks)

The diagram shows cuboid $ABCDEFGH$.

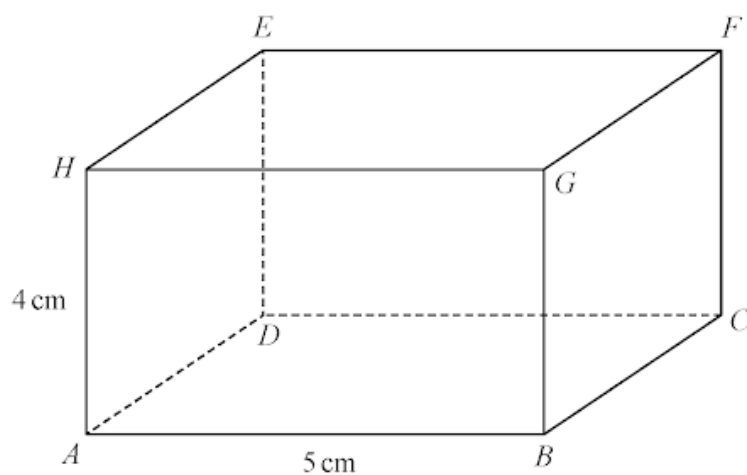


Diagram **NOT**
accurately drawn

$$AB = 5 \text{ cm}$$

$$AH = 4 \text{ cm}$$

The size of the angle between CH and the plane $ABCD$ is 35°

Calculate the volume of the cuboid.

Give your answer correct to 3 significant figures.

..... cm^3

(Total for Question is 5 marks)

The diagram shows a solid cylinder with radius 3 m.

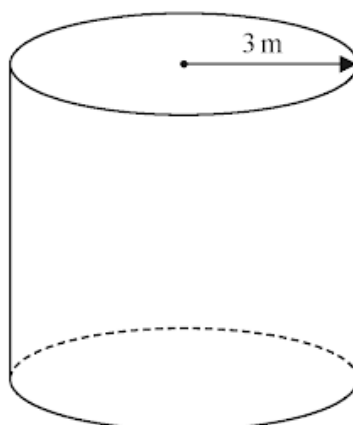


Diagram **NOT**
accurately drawn

The volume of the cylinder is $72\pi \text{ m}^3$

Calculate the **total** surface area of the cylinder.
Give your answer correct to 3 significant figures.

..... m^2

(Total for Question is 5 marks)

The diagram shows an isosceles triangle.

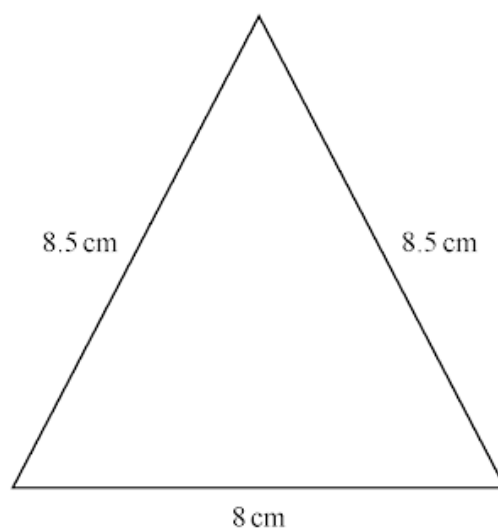


Diagram **NOT**
accurately drawn

Work out the area of the triangle.

.....cm²

(Total for Question is 4 marks)

The diagram shows a right-angled triangle.

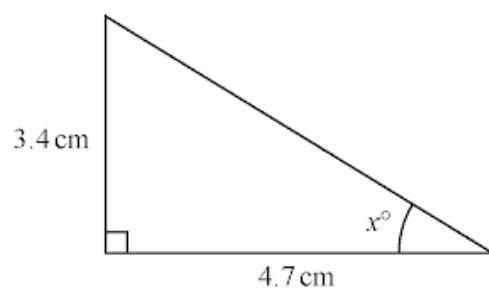


Diagram **NOT**
accurately drawn

Calculate the value of x .

Give your answer correct to one decimal place.

$x =$

(Total for Question is 3 marks)

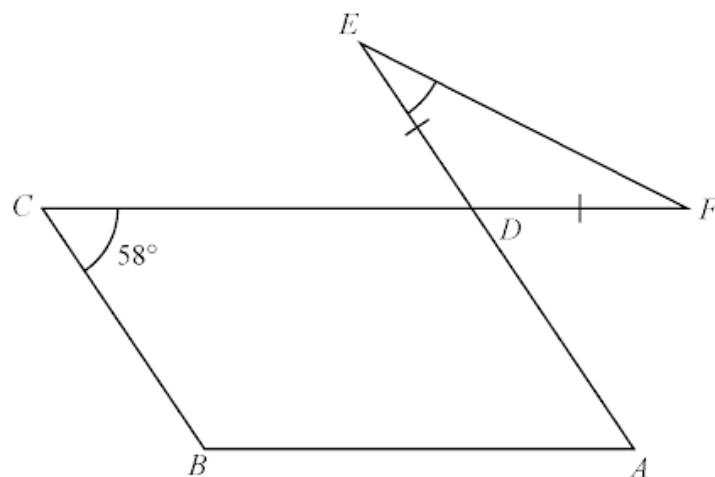


Diagram **NOT**
accurately drawn

The diagram shows a parallelogram $ABCD$ and an isosceles triangle DEF in which $DE = DF$

CDF and ADE are straight lines.

Angle $BCD = 58^\circ$

Work out the size of angle DEF .

Give a reason for each stage of your working.

○

(Total for Question is 5 marks)
